

Electric Submersible Pump Systems Course

(Day 2)

ESP Basic Design and Operational Factors

Course agenda

Day 1

**Overview of Artificial Lift Technology and
Introduction to ESP Systems**

Day 2

ESP Basic Design and Operational Factors

Days 3

ESP System Components and their Operational Features

Day 4

ESP System design: step-by-step procedure

Day 5

**ESP Installation Monitoring, Optimization, Troubleshooting
and Diagnostic**

ESP Basic Design and Operational Factors

- ✓ Basic concepts related to the centrifugal pumps
- ✓ Conceptual ESP's design
- ✓ Total Dynamic Head (TDH)
 - Well-Head Tubing Pressure (HTHP)
 - Net Vertical Lift (NLV)
 - Total Friction Loss
- ✓ ESP basic design: Step by step procedure
- ✓ Workshop Session: ESP basic design
- ✓ Operational Factors affecting the ESP performance
 - Axial Thrust Forces
 - Fixed vs Floating Impellers
 - Axial forces compensation
 - Cavitation
- ✓ Electricity and Magnetism: Review of Fundamentals

ESP basic design

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**Basic concepts
related to the
centrifugal pumps**

ESP design

Basic concepts related to the centrifugal pumps

At the end of this section, you will be able to understand:

- The physic governing a centrifugal pump
- How a centrifugal pump work
- The Head concept and how the centrifugal pump generates the Dynamic Head
- The relation among Pump Head and fluid rate
- How a centrifugal pump can be regulated

An introduction to Centrifugal Pumps

ESP design

Basic concepts related to the centrifugal pumps

Centrifugal Pumps

A centrifugal pump, **according the Bernoulli Equation, converts the input power (electric) to kinetic energy** in the liquid by accelerating the liquid by a revolving device (the impeller) and then **into pressure energy or Dynamic Head.**

The faster the impeller revolves or the bigger the impeller is, the higher will the velocity of the liquid energy transferred to the liquid be. This is described by the **Affinity Laws.**

Basic concepts related to the centrifugal pumps

Bernoulli Equation

As already mentioned, a centrifugal pump **converts the input power (electric) to kinetic energy** in the liquid by accelerating the liquid by a revolving device, an impeller, according the **Bernoulli Equation**.

The **Bernoulli Equation** states:

*For a Newtonian,
incompressible fluid,
in steady flow,
the sum of Kinetic energy,
Pressure energy and
Potential energy,
per unit volume
is constant at any point*

$$\frac{\mathbf{v}^2}{2\mathbf{g}} + \frac{\mathbf{P}}{\rho\mathbf{g}} + \mathbf{H} = \text{Constant}$$

where

$\mathbf{v}$ = flow speed

$\mathbf{P}$ = pressure

ρ = density

$\mathbf{g}$ = gravity

$\mathbf{H}$ = height

*This equation is often referred to the **head** because all elements has the unit of length.*

Basic concepts related to the centrifugal pumps

Pressure and Head

If the discharge of a centrifugal pump is pointed straight up into the air (no impeller – no flow obstruction – no flow resistance) *the fluid will pumped to a certain height - or **head*** - called the shut off head.

This maximum head *is mainly determined by the outside diameter of the pump's impeller and the speed of the rotating shaft*. The head will change as the capacity of the pump is altered.

The kinetic energy of a liquid coming out of an impeller is obstructed by creating a resistance in the flow. The first resistance is created by the pump casing which catches the liquid and slows it down. *When the liquid slows down the kinetic energy is converted to pressure energy.*

It is the resistance to the pump's flow that is read on a pressure gauge attached to the discharge line

A pump does not create pressure, it only creates flow. Pressure is a measurement of the resistance to flow.

Basic concepts related to the centrifugal pumps

Pressure and Head

In Newtonian fluids (non-viscous liquids like water or gasoline) ***the term head is used to measure the kinetic energy which a pump creates.***

Head is a measurement of the height of the liquid column the pump creates from the kinetic energy the pump gives to the liquid.

The main reason for using ***head instead of pressure*** to measure a centrifugal pump's energy ***is that the pressure from a pump will change if the specific gravity (weight) of the liquid changes, but the head will not***

It is important to understand that the centrifugal pump will pump all fluids to the same height if the shaft is turning at the same rpm

The only difference between the fluids is the amount of power it takes to get the shaft to the proper rpm: the higher the specific gravity of the fluid the more power is required.

So, the required power to achieve a certain fluid pressure - or head - is proportional to the specific gravity of the fluid.

Basic concepts related to the centrifugal pumps

Pressure-Head

From Bernoulli Equation:

$$\frac{v_1^2}{2g} + \frac{P_1}{\rho g} + H_1 = \frac{v_2^2}{2g} + \frac{P_2}{\rho g} + H_2$$

where

v_1 = inlet flow speed

v_2 = outlet flow speed

P_1 = inlet pressure

P_2 = outlet pressure

H_1 = inlet height

H_2 = outlet height

ρ = density

g = gravity

If:

$v_1 = 0$; $H_1 = 0$; $P_1 = P_2$ (no obstruction)

$$H = \frac{v_2^2}{2g}$$

and

$$v = \frac{N \times D}{229}$$

where:

H = Total head developed in ft

v = Velocity at periphery of impeller in ft/sec

g = Acceleration due to gravity – 32.3 feet/sec²

N = The impeller RPM (revolutions per minute)

D = Impeller diameter in inches

Basic concepts related to the centrifugal pumps

Example of Head calculation

Data

D = Impeller diameter: 5 inches

N = Impeller RPM: 3000 revolutions per minute

g = Acceleration due to gravity – 32.3 feet/sec²

Calculate Head developed by one single stage:

$$v = \frac{N \times D}{229} \quad v = (3000 \times 5) / 229 = 66 \text{ ft/sec}$$

$$H = \frac{v^2}{2g} \quad H = (66)^2 / (2 \times 32.3) = 66 \text{ ft}$$

Convert the head in pressure supposing that the pumped fluid is freshwater :

P = pressure psi

G_w = freshwater gradient = 0.433 psi/ft

P = H x G_w = 66 x 0.433 = 28.58 psi

An introduction to Discharge Regulation of Centrifugal Pumps

Basic concepts related to the centrifugal pumps

Discharge Regulation of Centrifugal pump

It is often necessary to adapt the pump capacity to a temporary or permanent change in the process demand.

The capacity of a centrifugal pump can be regulated either at

- constant speed, or
- varying speed

Basic concepts related to the centrifugal pumps

Capacity Regulation by Constant Speed

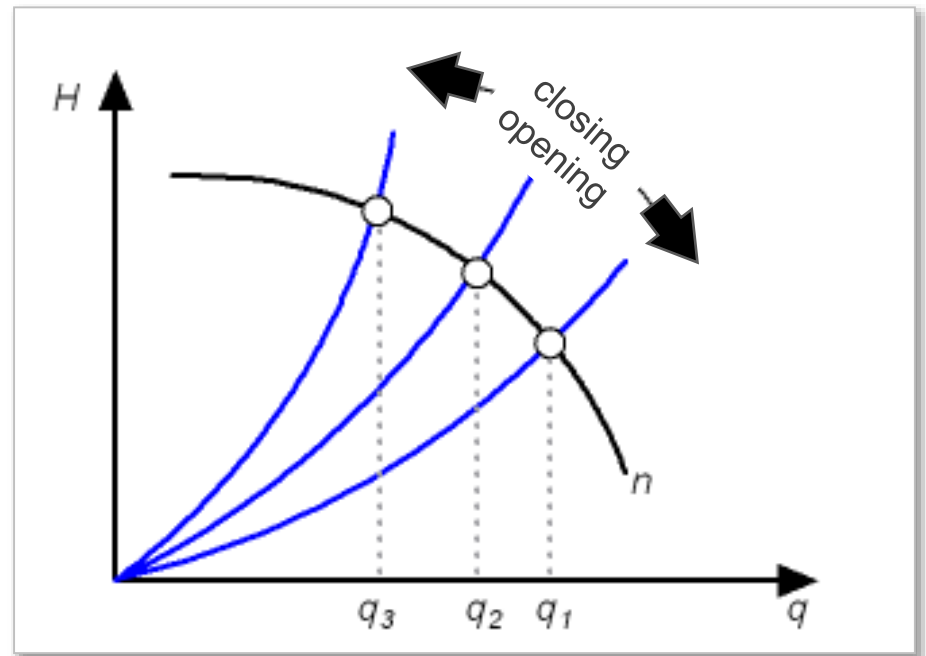
Capacity can be regulated at constant speed by

- throttling
- bypassing flow
- changing impeller diameter
- modifying the impeller

Throttling

Having the ESP at bottom-hole, the only possible way **to change the rate by constant speed** is by throttling through surface Choke variation, unless decision to perform a work-over is taken (e.g. ESP down or upgrading by changing motor, type and number of stages).

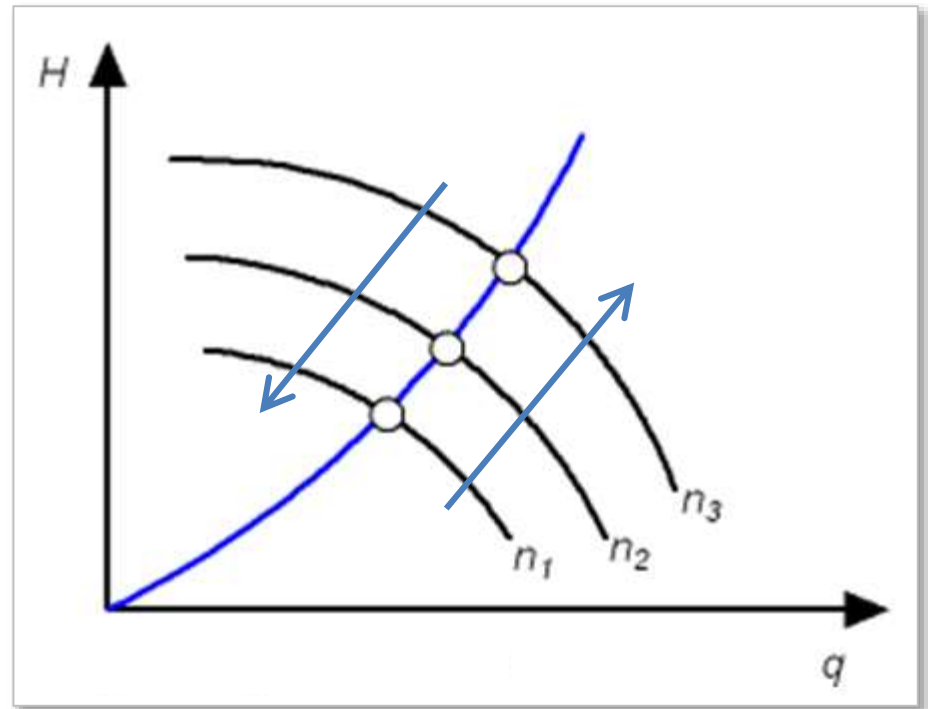
Throttling through Choke reduction is energy inefficient since the energy to the pump is not reduced. **Energy is simply wasted by increasing the dynamic loss.**



Basic concepts related to the centrifugal pumps

Capacity Regulating by Varying Speed

- The **speed of the pump can be varied with variable speed drives** - inverters - AC drives - adjustable frequency drives - operates by varying the frequency and voltage to the electric motor.
- **Speed regulating is energy efficient** since the energy to the pump is reduced with the decrease of speed.
- The change in power consumption, head and volume rate can be estimated with the **affinity laws**.



Conceptual ESP's design

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Basic concepts for ESP selection

ESP conceptual design

At the end of this section, you will be able to:

- Explain the purpose of a pump (any pump) in a well.
- Understand the Basic concepts for ESP selection

and:

- Determine the pump suction and discharge pressure
- Calculate the required Total Pump-Head
- Choose the number of stages that a given centrifugal stage type pump requires to meet a flow target
- Determine the power requirements to run the pump
- Size a pump using the TDH requirement

ESP conceptual design

Basic concepts for ESP selection

Sizing an oil-well pump comprises two primary duties:

- 1. Determining the required pumping pressure***
- 2. Selecting a pump to fulfil the pumping requirements***

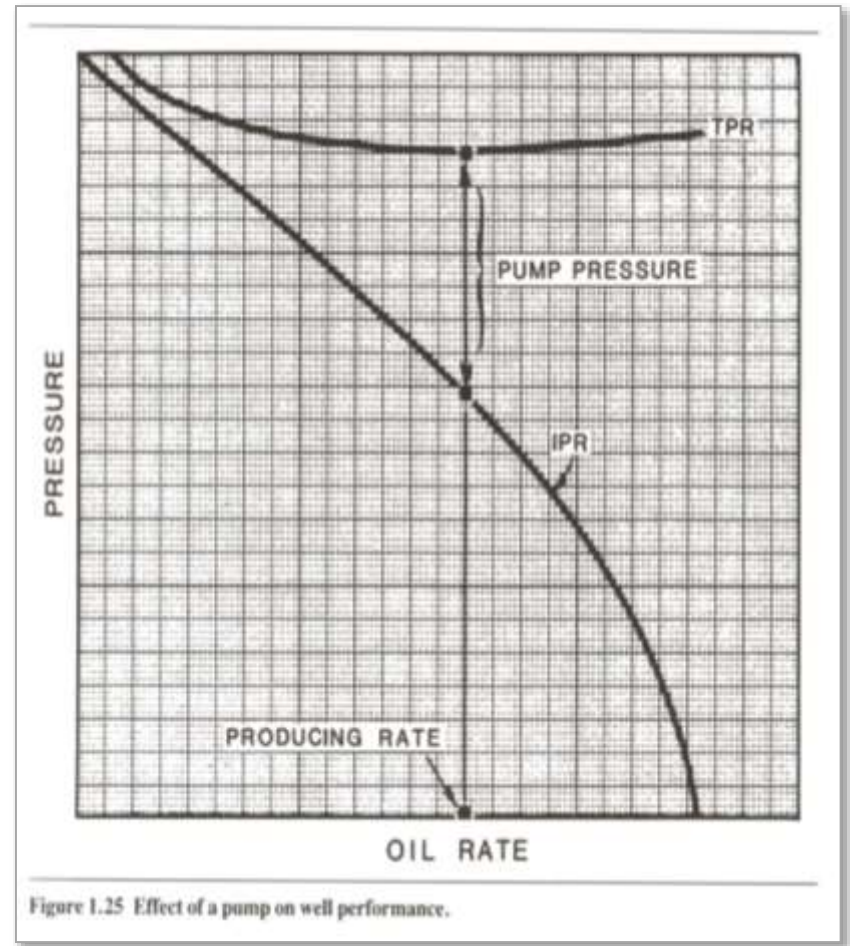
The pumping requirement is merely ***the pumping pressure needed to maintain a desired wellbore flowing pressure or a desired production rate.***

ESP conceptual design

Required pumping pressure

The **required pumping pressure** to produce a desired fluid rate Q is determined by combining:

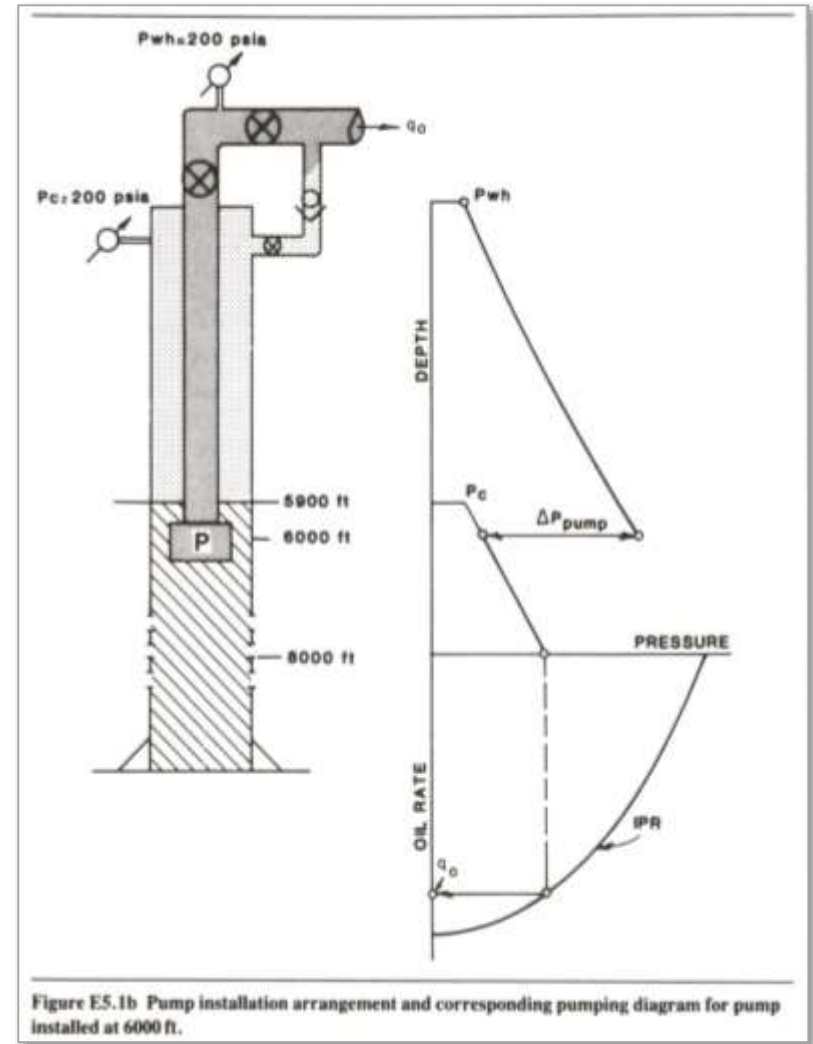
1. *The well's inflow performance (IPR)*
2. *The tubing performance curve (TPR).*



ESP conceptual design

This figure provides (graphically) the **conceptual procedure for establishing the required pumping pressure**:

1. The IPR provides the P_{wf} corresponding at a selected oil rate Q_o and vice versa
2. Through the casing flow gradient the **pump suction pressure** can be estimated
3. Imposing a wellhead flowing pressure P_{wh} , the Tubing Flowing Gradient (at a selected oil rate Q_o) provides the Flowing TBG pressure at pump depth corresponding to the **pump discharge pressure**.
4. The difference among **discharge pressure and the pump suction pressure gives the ΔP to be provided by pump**.



ESP conceptual design

Pumps providing the pressure needed to boost production

Simply enough, a pump – **any pump** – provides the pressure required by the well to match a given production rate.

Different pumps do this in different ways:

- **Centrifugal pumps** have a distinct pressure/flow profile for each type of stage.
- **Sucker Rod Pumps** have a fixed flow rate for a given design (stroke, speed, plunger diameter)
- **PCPs**, like rod pumps, have a fixed flow rate for a given design (rotor/stator design, rotational speed)
- **Screw pumps** have a fixed flow rate for a given design (screw design, rotational speed)

ESP conceptual design

Centrifugal pump performance curves

The **characteristics curves for centrifugal pumps** are reported at constant driving speed:

- 3500 RPM with 60 Hertz AC electrical supply (typical in USA)
- 2915 RPM with 50 Hertz AC electrical supply (typical in Europe)

In most pumping applications the electrical power supply to the driving motor has a **constant frequency** and thus the pump is operated at **constant speed**.

The **constant-speed characteristic curve** therefore provides all the necessary information on pump pressure and power requirements.

ESP conceptual design

Pump performance curves

Determining the **actual pressure-rate-power relationship** from the water-related pump characteristic curve require the following procedure:

1. Determine the **pump setting depth**.
2. Determine the **required pumping pressure** to produce a desired fluid rate Q by combining the IPR with TPR.
3. Convert the required pumping pressure **to pump-head**:

$$H = \frac{\Delta P}{\rho} = \frac{\Delta P}{0.433\gamma} = 2.31 \frac{\Delta P}{\gamma}$$

where:

ρ is density of pumped fluid in lb/ft³

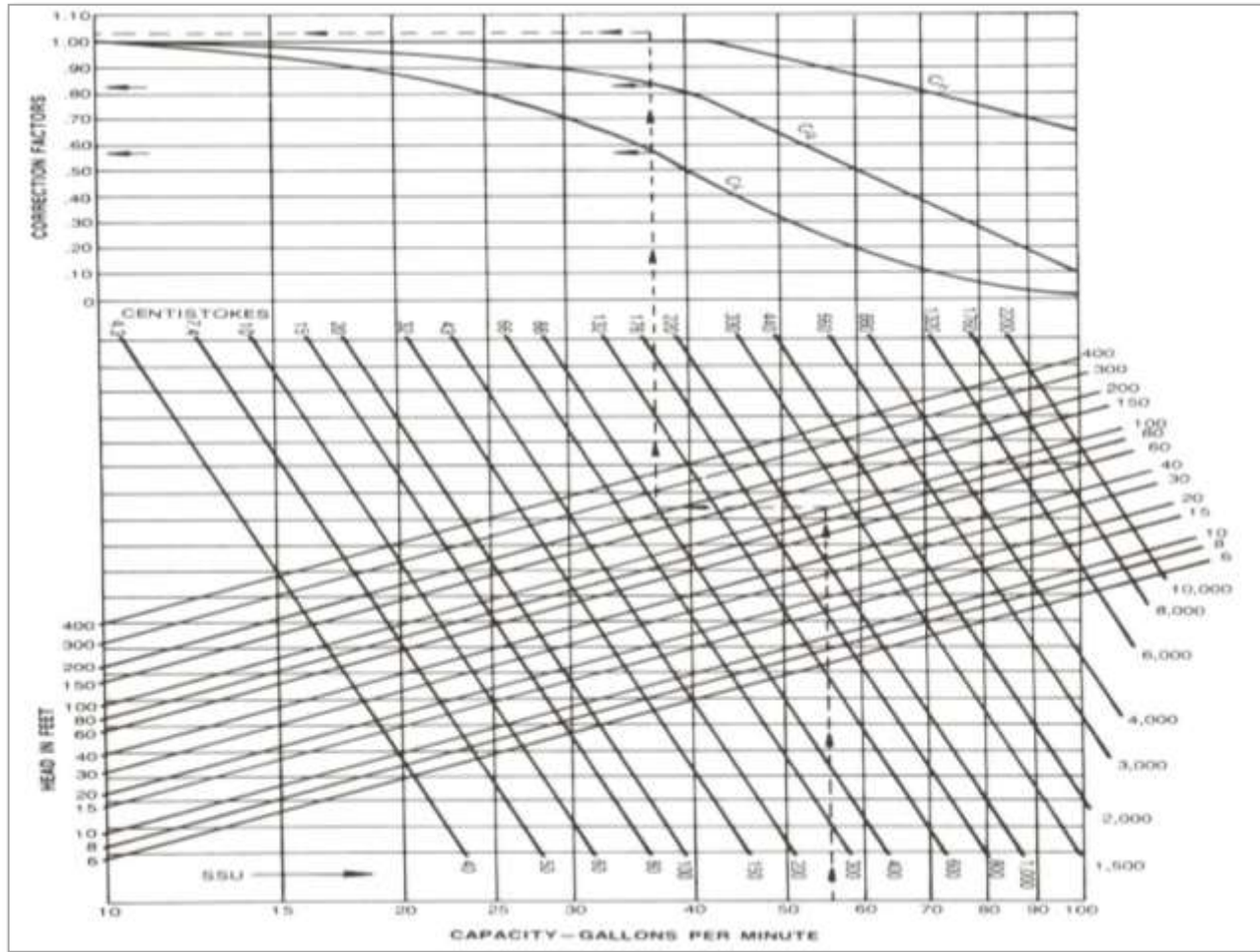
γ is specific gravity of pumped fluid (water=1)

4. Correct the reported pumping head for actual liquid viscosity if substantially different than 1 cp (e.g. make use of American Hydraulic Institute)
5. Correct the **power requirements** for the density of pumped fluid:

$$W = W_{\text{water}} \times SG_{\text{produced fluid}}$$

ESP conceptual design

Viscosity correction factors for the density of pumped fluid



1. Enter the required fluid rate (capacity)
1 GPM = 34.56 bpd
1 bpd = 0.29 GPM

2. Move vertically to encroach the proper single stage Head line

3. Move horizontally to encroach the proper viscosity line provided in SSU and centistokes
 $cp = \text{centistokes} \times SG$

4. Move vertically to encroach the Correction Factors (C_E , C_Q , C_H) lines

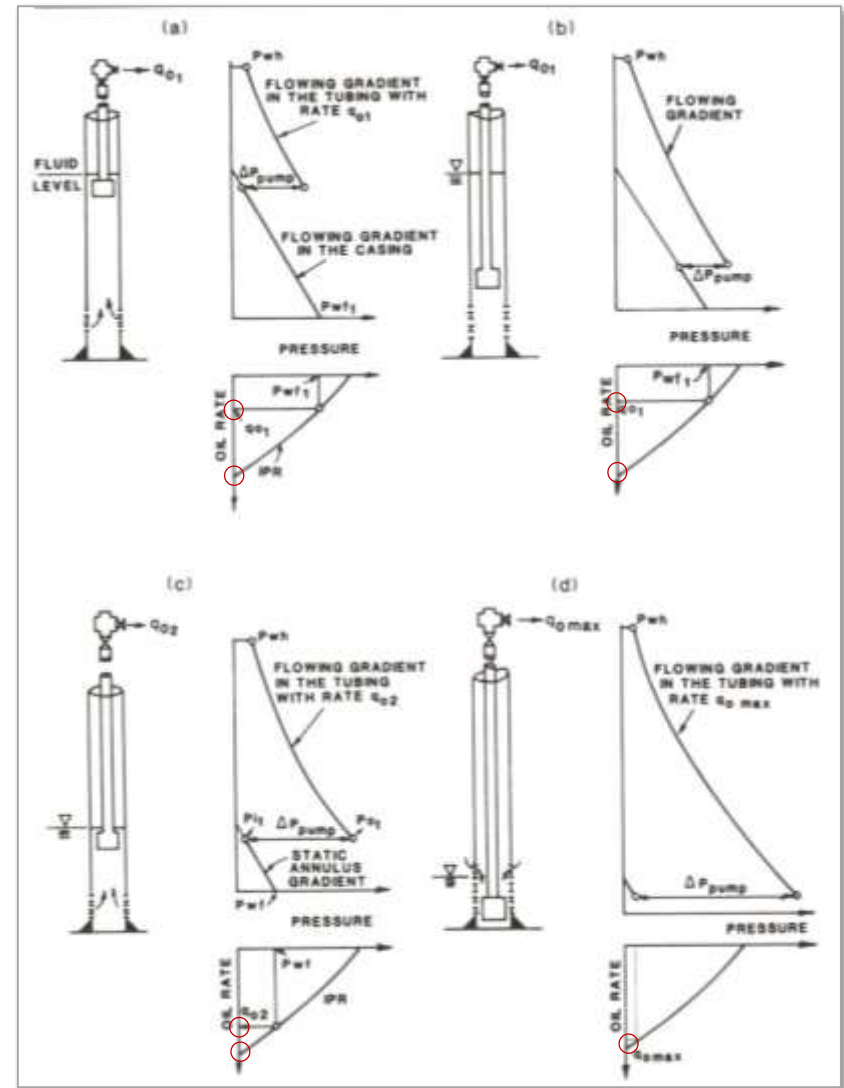
ESP conceptual design

Pump setting depth and size

Two extreme pumping conditions are considered in figure:

1. Figure (a) and (b) illustrates cases where the pump rate is much lower than the absolute open flow. This situation typically occurs when an **undersized pump is used**. Undersized pump is usually operated at maximum capacity.
2. The other extreme case is a well pumping near its maximum well capacity, as in fig (c) and (d). In this case **the inflow performance limits the possible production rate and not the lift system**.

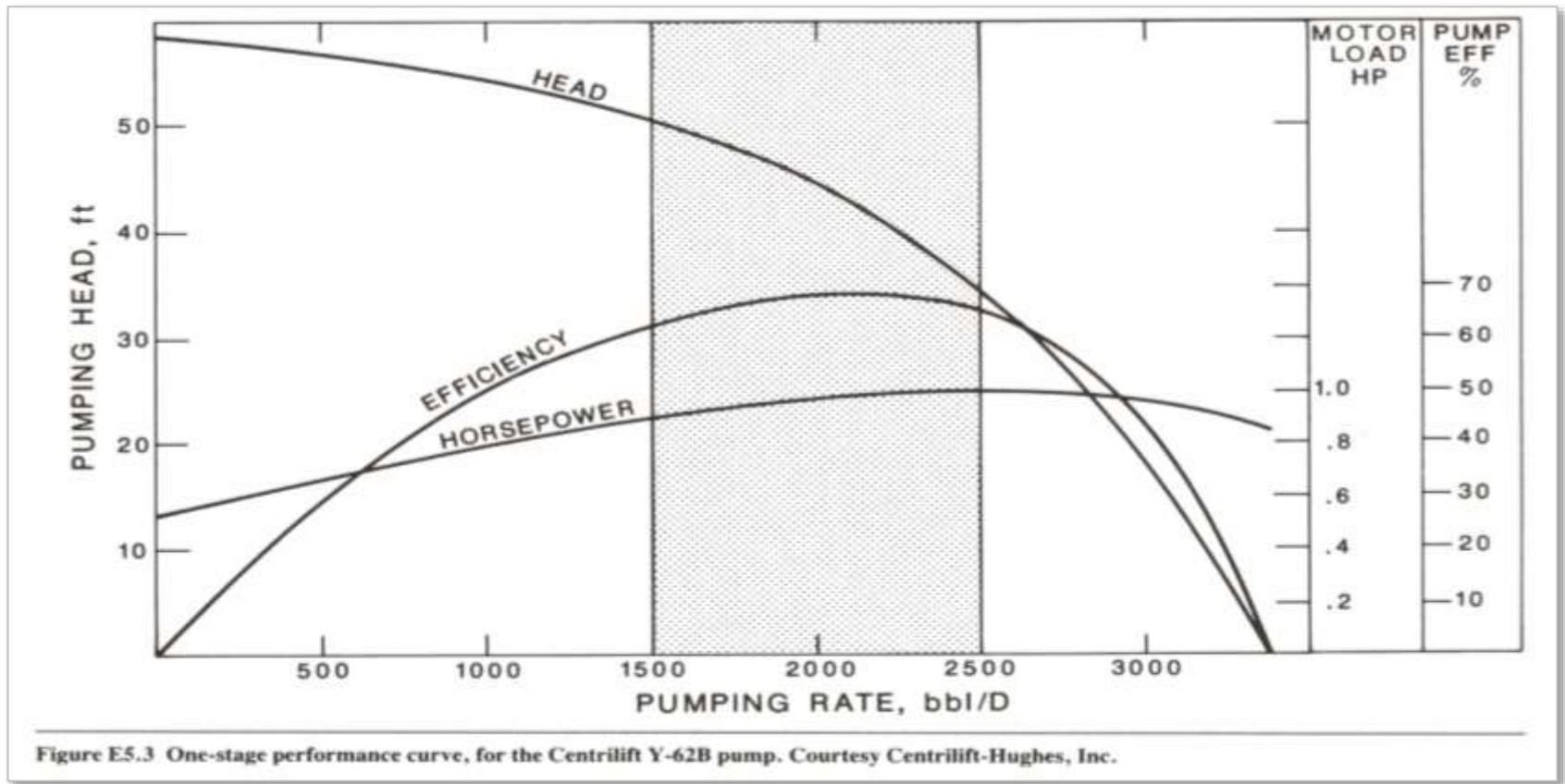
A general observation is that **the pumping rate increases with pump setting depth** (e.g. below the perforations). This because a **minimum backpressure** is applied and an **efficient gas separation** is achieved.



ESP conceptual design

Centrifugal pump performance curve

Pump manufacturers publish **pump performance curves** that describe **head/flow-rate relationship**. These curves are typically published **for one stage**



ESP conceptual design

Multistage Centrifugal Pumps

- Centrifugal pumps are built with multiple stages in SERIES.
- The outlet of the lower stage becomes the inlet of the upper stage.
- Therefore, the pressure is additive across the stages.
- If we assume that we have the same flow rate through each stage, then *the **total pressure** (P_{tot}) the pump produces at a given flow rate is equal to the **pressure that one stage produces** (P_{stage}) multiplied by the **number of stages** (N):*

$$P_{tot} = P_{stage} \times N$$

ESP conceptual design

Centrifugal Pumps Hydraulic horsepower

- The actual work done by the pump (the hydraulic horsepower or hhp) on the fluid is.

$$\text{hhp} = 1.7 \times 10^{-5} \Delta P Q$$

were:

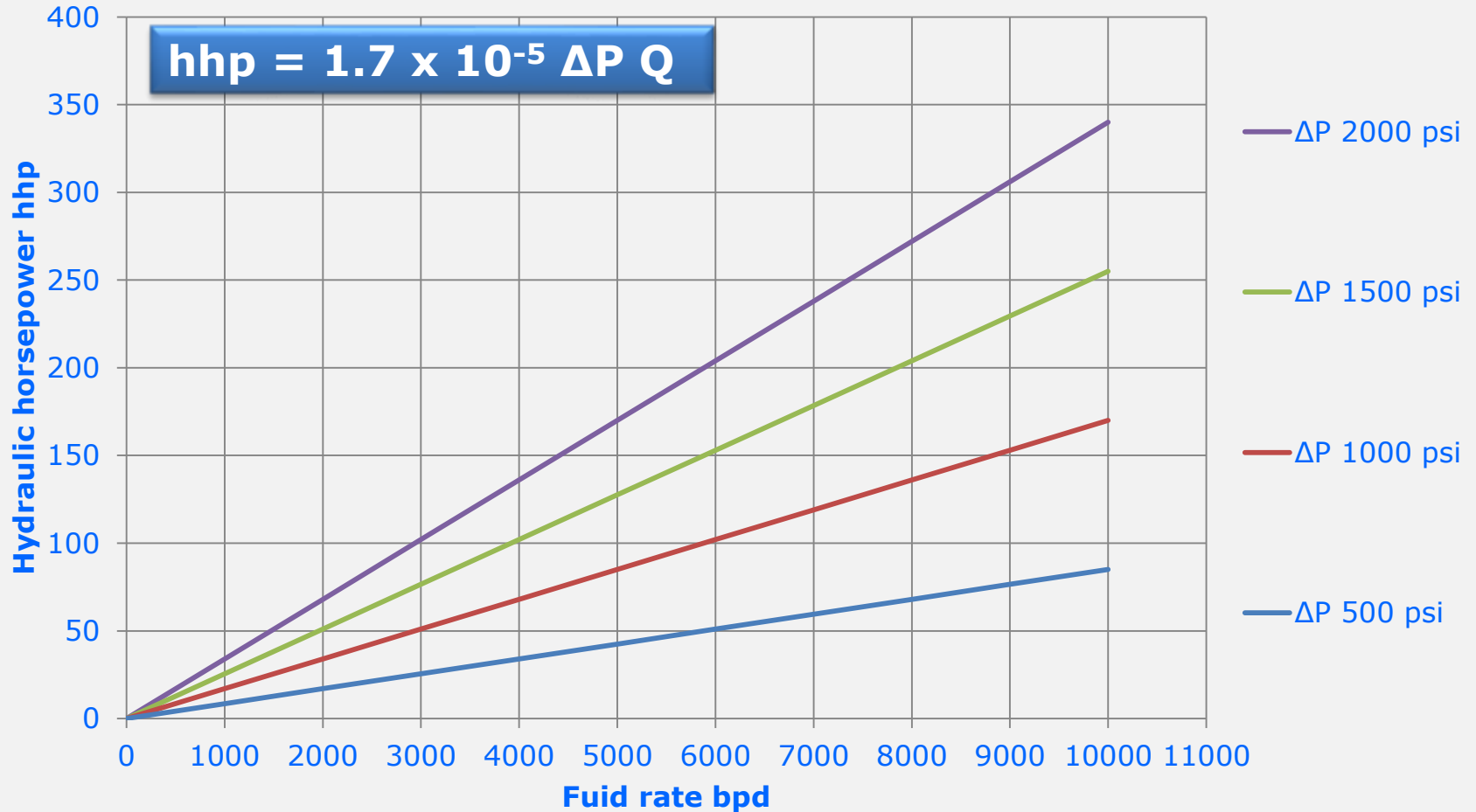
ΔP is the pressure difference across the pump (psi)

Q is the flow rate (bpd)

- So, the pumping power is proportional to the differential pressure ΔP and flow rate Q .

ESP conceptual design

Hydraulic horsepower vs Fluid rate



Total Dynamic Head (TDH)

ESP design: Total Dynamic Head

At the end of this section, you will be able to:

- Calculate the TDH by breaking it down into its three components:
 - H_{THP}
 - Net Vertical Lift (NVL)
 - H_{friction}
- Explain the concept of NVL compared to true fluid level
- Convert from TDH to Pressure and vice-versa

Calculation of Total Dynamic Head (TDH)

ESP design: Total Dynamic Head

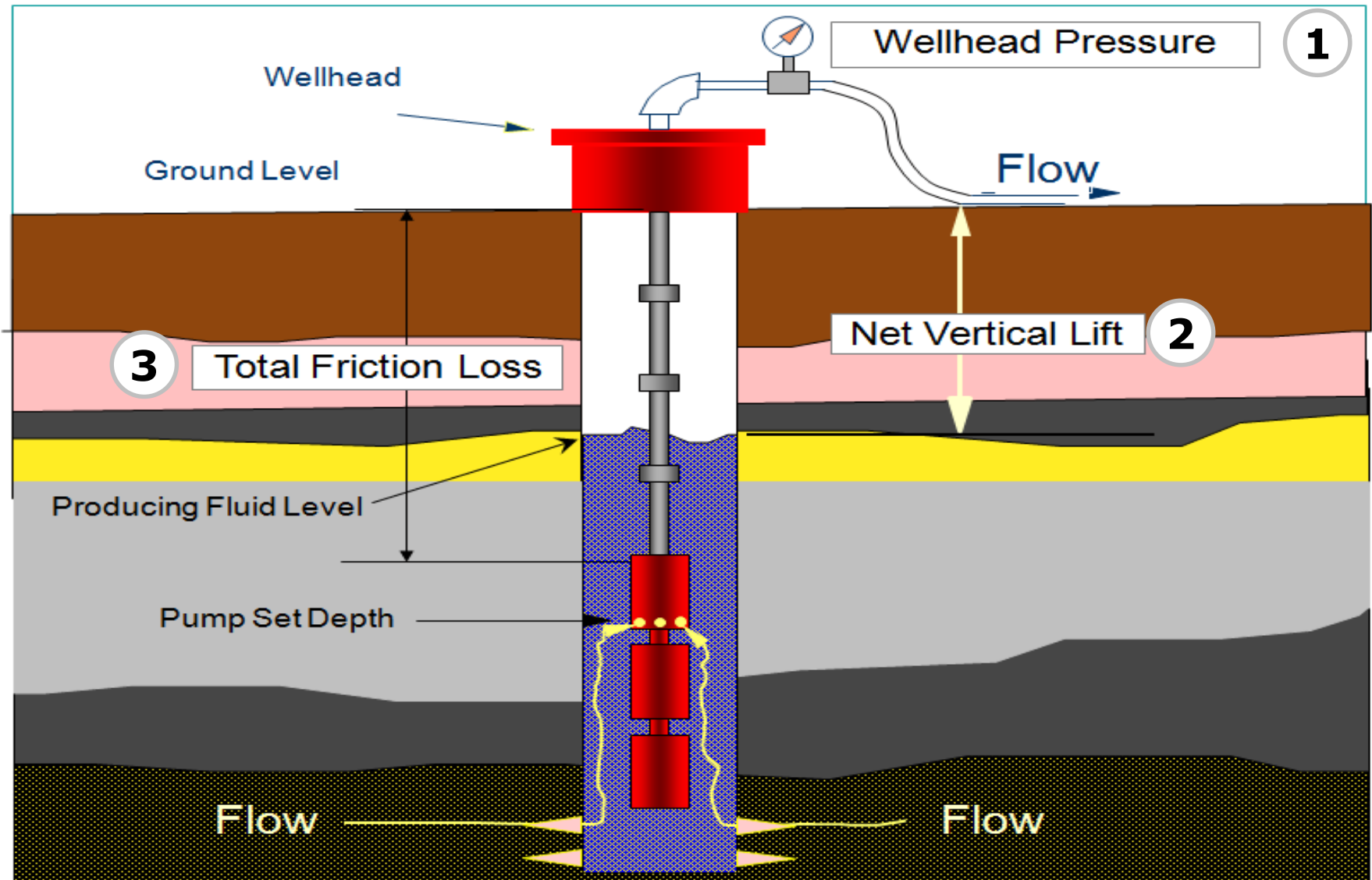
TDH is the sum of three basic components:

1. the **Well-Head Tubing Pressure (H_{THP})** at a given liquid production rate, which acts as backpressure.
2. the **net hydrostatic pressure** acting on the pump, named **Net Vertical Lift (NVL)** referring to the vertical distance (elevation) which the fluid must be lifted.
3. the **frictional pressure drop** that occurs in the tubing at a given liquid rate, named **Total Friction Loss (H_{friction})**.

$$\text{TDH} = H_{\text{THP}} + \text{NVL} + H_{\text{fr}}$$

ESP design: Total Dynamic Head

The TDH's three components



Well-Head Tubing Pressure (H_{THP})

ESP design: Head Tubing Pressure

Well-Head Tubing Pressure

Well-Head Tubing Pressure is sometimes called "Surface Pressure", "Back Pressure" or even "Flow-line Pressure". Actually the most accurate term is "Tubing Discharge Pressure" since this is the pressure at the discharge of the tubing from the well.

- For the purposes of this example, we will assume that the Well-Head Tubing Pressure (THP) is 500 psi.
- To convert the THP in **HEAD (H_{THP})**:

$$H = 2.31 \frac{P}{\gamma} = 2.31 \frac{THP}{\gamma} = H_{THP}$$

Where γ is the specific gravity of pumped fluid (water=1)

If $\gamma = 0.8$

$$H_{THP} = 2.31 \times 500 / 0.8 = 1444 \text{ ft}$$

Net Vertical Lift

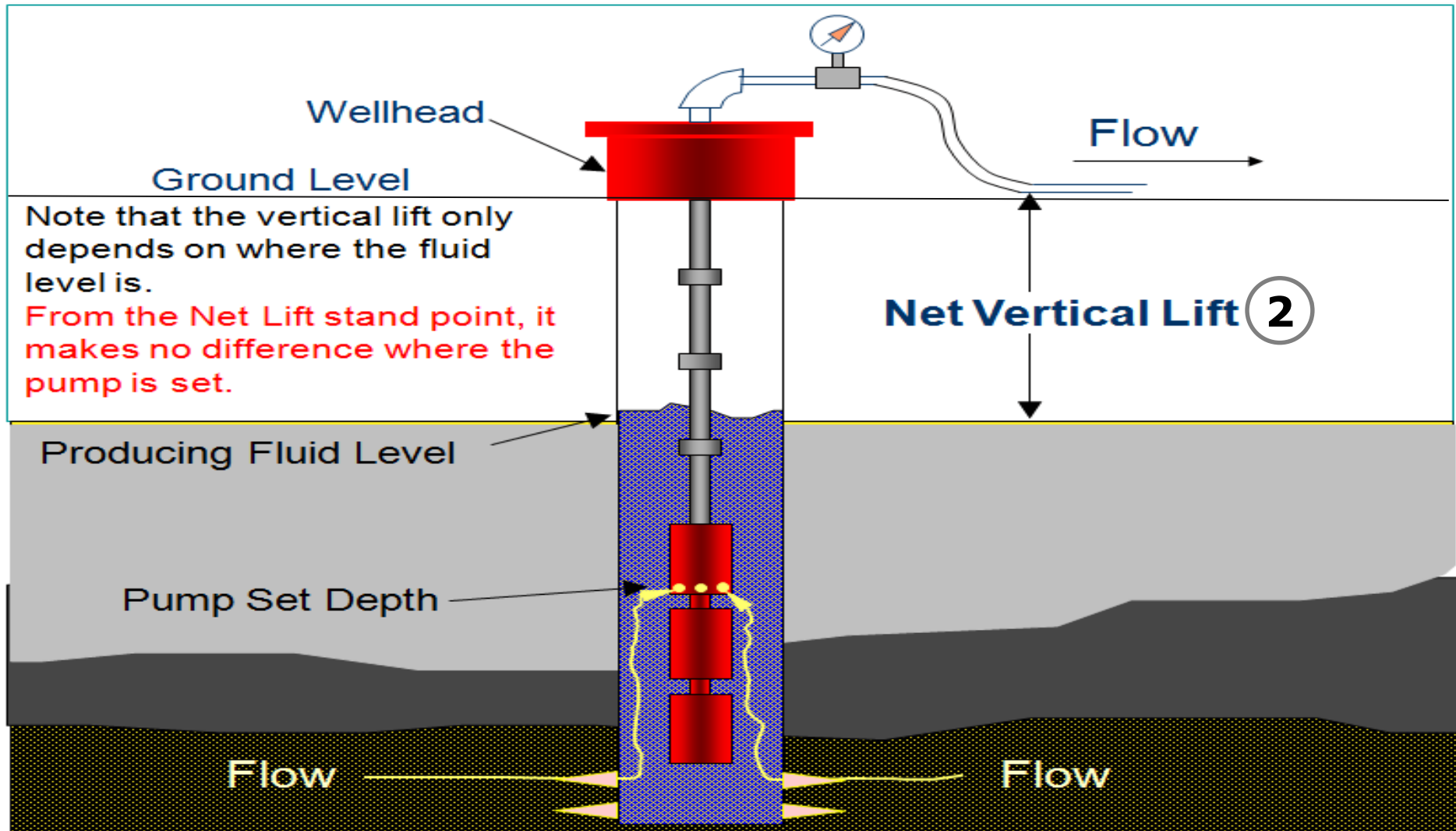
NVL

ESP design: Net Vertical Lift

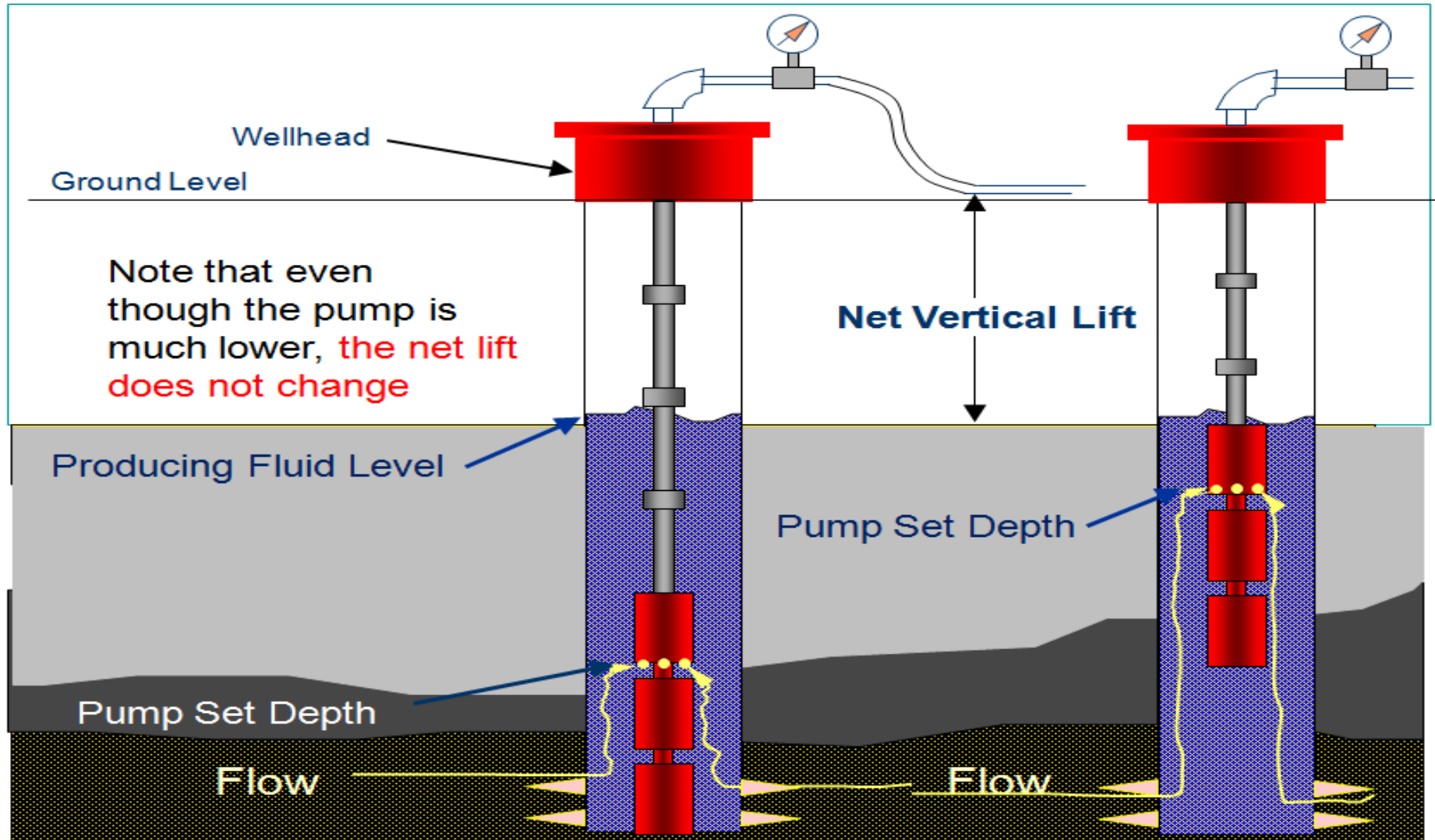
Net vertical lift

- The **Net Vertical Lift (NVL)** is the **vertical distance through which the fluid must be lifted** to get to the surface.
- Regardless of where the pump is set, or well inclination, the Net Vertical Lift will NOT change.

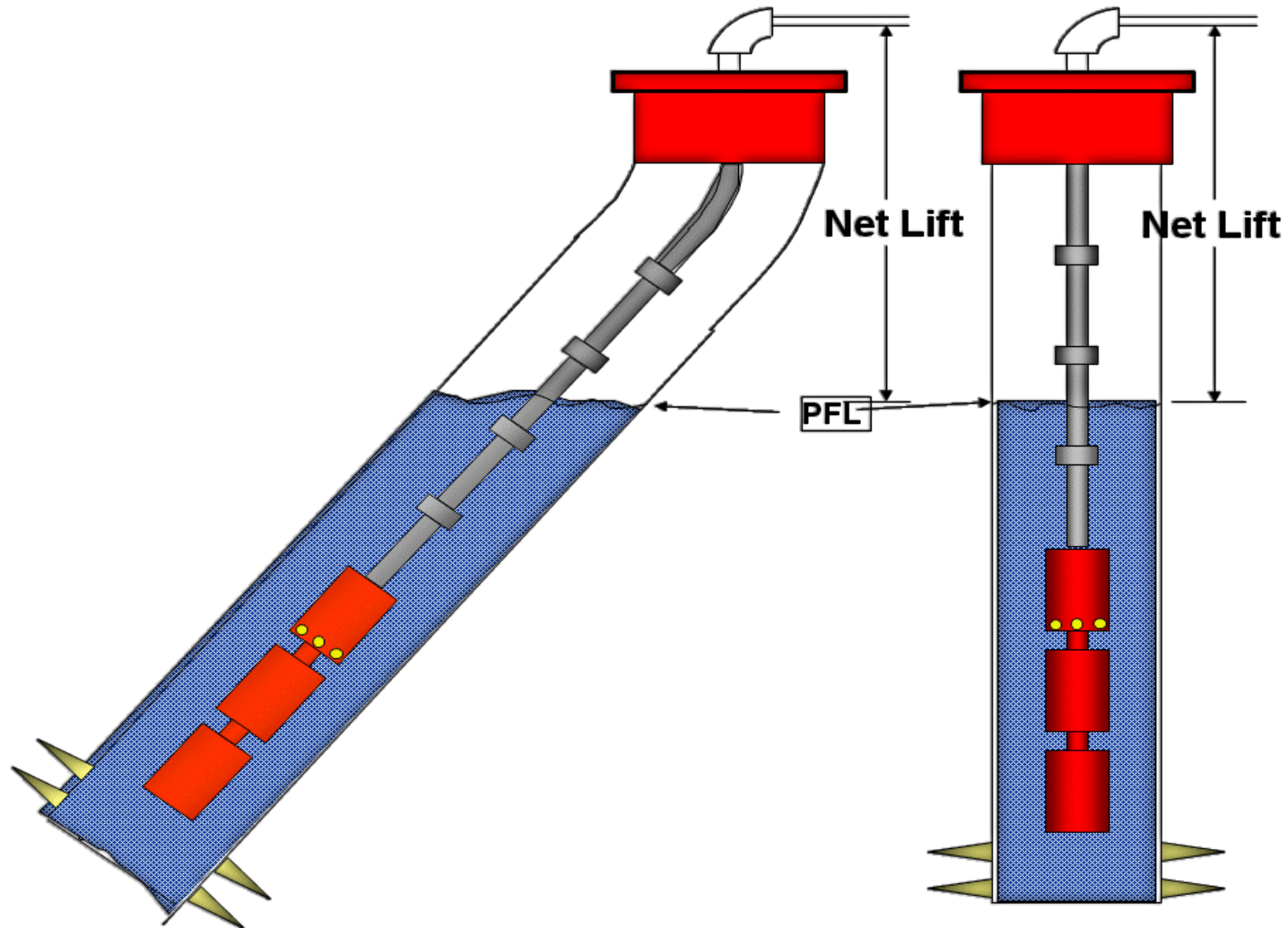
ESP design: Net Vertical Lift



ESP design: Net Vertical Lift



ESP design: Net Vertical Lift



ESP design: Net Vertical Lift

Net vertical lift

As already mentioned, the **net vertical lift** is the hydrostatic pressure acting on the pump.

From another point, **net vertical lift** is also the **vertical distance through which the fluid must be lifted by the pump**, that is its **Head**.

For the purposes of this example, we will assume we are given a fluid level of 4000 feet from surface (vertical distance).

$$\text{Net Vertical Lift} = 4000 \text{ ft}$$

Total Friction Loss

ESP design: Friction Loss

Friction or Dynamic Loss

Friction or Dynamic loss is an energy loss (we actually measure it as a pressure loss) due to viscous shear of the flowing fluid.

The Friction Pressure determination is a relatively easy task when we are dealing with single-phase liquid, but is a complex issue in case of complex fluid mixtures.

For frictional pressure drop calculation, make reference to Well Deliverability section where this topic is discussed.

ESP design: Friction Loss

Total Friction or Dynamic Loss

- For the purposes of this example, we will assume that the Friction losses to move the producing fluid from the pump depth to the surface, at operating pump rate, have been estimated in 250 psi.
- To convert the pressure friction losses in **friction HEAD**:

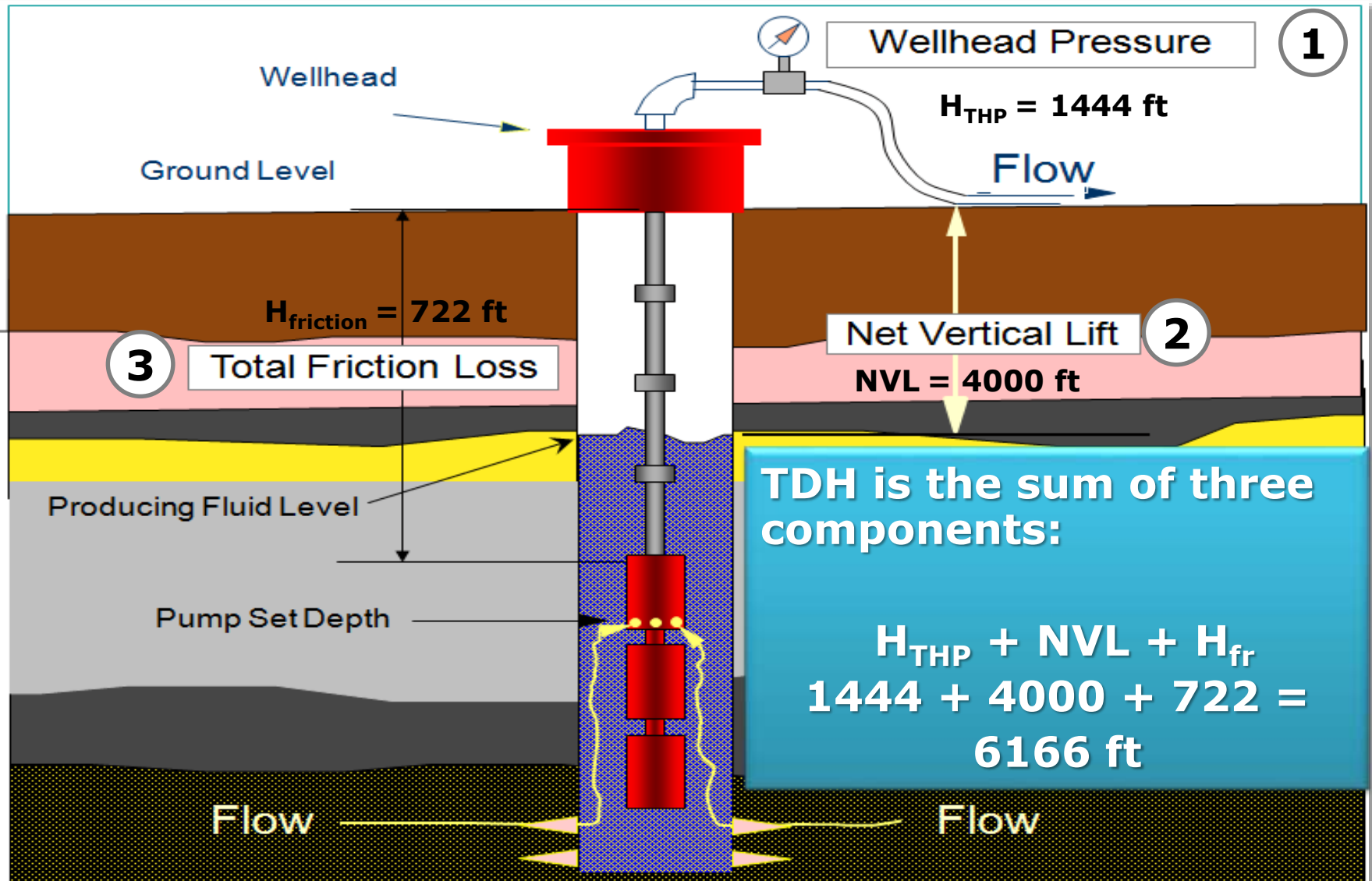
$$\mathbf{H} = 2.31 \frac{\mathbf{P}}{\gamma} = 2.31 \frac{\mathbf{P}_{friction}}{\gamma} = \mathbf{H}_{fr}$$

where γ is the specific gravity of pumped fluid (water=1)

If $\gamma = 0.8$

$$\mathbf{H}_{fr} = 2.31 \times 250 / 0.8 = 722 \text{ ft}$$

ESP design: Total Dynamic Head



ESP basic design - step-by-step procedure

Main source: Petroleum Production Engineering. A Computer-Assisted Approach B. Guo, W.C. Lions, A. Ghalambor - Elsevier

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ESP design: Step-by-step procedure

Factors affecting the ESP selection

The following factors are important in designing ESP applications :

1. PI of the well
2. Casing and tubing size
3. Static liquid level.

ESP are usually for high PI.

The outside diameter (OD) of the ESP is determined by the minimum inside diameter (ID) of the borehole.

There must be clearance around the outside of the pump to allow the free flow of produced fluid to the pump intake.

ESP design: Step-by-step procedure

Conceptual ESP design procedure

The following procedure can be used for the ESP selection:

1. Starting from well inflow performance relationship (IPR), determine a **desirable liquid production rate Q_{LP}**
2. Select a pump size from the manufacturer's specification that has **minimum deliverability flow rate Q_{LD} , that is, $Q_{LD} > Q_{LP}$**
3. From the IPR, determine the flowing bottom-hole pressure **P_{wf} at the pump-delivering flow rate Q_{LD} (not the Q_{LP}).**

Remember: if the reservoir is not capable to feed the pump (due to low PI, high Skin, strong depletion, etc....) the pump cannot give us any oil !!!!

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ESP design: Step-by-step procedure

4. Assuming zero casing pressure and neglecting gas weight in the annulus, calculate the minimum pump depth **(to be sure that the pump is submerged in the producing fluid)** by:

$$D_{\text{pump}} = D - (P_{\text{wf}} - P_{\text{suction}}) / 0.433SG_L$$

where:

D_{pump}	= minimum pump dept, ft
D	= depth of producing interval, ft
P_{wf}	= flowing bottom-hole pressure, psi
P_{suction}	= required suction pressure of pump, 150-300 psi
SG_L	= specific gravity of producing fluid (1 freshwater)

The above equation is derived from pressure balance equation:

$$P_{\text{wf}} = (D - D_{\text{pum}}) \times (0.433SG_L) + P_{\text{suction}}$$

The P_{wf} is balanced by hydrostatic column $(D - D_{\text{pum}})SG_L$, and the required suction pressure P_{suction} .

ESP design: Step-by-step procedure

5. Determine the **required pump discharge pressure** based on **wellhead flowing pressure and tubing flow gradient at pump-delivering flow rate Q_{Lp}** . This can be carried out using a dedicated computer program, derived from a proper Kermit E. Brown Flowing Pressure Gradient working graph if available, or by Hazen-Williams approach (NO free gas).

6. Calculate the **required pump pressure differential**

$$\Delta P = P_{\text{discharge}} - P_{\text{suction}}$$

$$P_{\text{discharge}} = 150 - 300 \text{ psi}$$

and then the **convert** the required pump pressure differential in **required pumping head (H)** by: **$H = 2.31(\Delta P / SG_L)$**

7. From the manufacturers' pump characteristics curve, **read pump head or head per stage**. Then **calculate the required number of stages**.
8. Determine the **total power required (W)** for the pump by multiplying the power per stage by the number of stages.
9. **Correct the total power required** for the density of pumped fluid

$$W = W_{\text{water}} \times SG_{\text{produced fluid}}$$

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ESP design: Step-by-step procedure

Example - ESP basic design

Problem

A 10.000 ft-depth well produces 32 °API (SG = 0.865) oil with GOR 50 scf/std and zero water cut through a 3-in (2.992-in ID) tubing in a 7-in casing.

The oil has a formation volume factor of 1.25 and average viscosity of 5 cp. Gas specific gravity is 0.7.

The surface and bottom-hole temperature are 70 °F and 170 °F, respectively.

The IPR of the well can be described by the Vogel model with a reservoir pressure **4.350 psi** and **AOF 15.000 stb/day**.

If the well is to be put in production with an ESP to produce liquid at 8.000 stb/day against a flowing wellhead pressure of 100 psi, determine the required specifications for an ESP for this application. Assume the minimum suction pressure is 200 psi.

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ESP design: Step-by-step procedure

Solution

1. Vogel's well IPR gives:

$$P_{wf} = 0.125P_R \left[\sqrt{81 - 80 \left(\frac{Q_o}{Q_{\max}} \right)} - 1 \right]$$

$$P_{wf} = (0.125 \times 4350) \times \{[81 - 80 \times (8000/15000)]^{1/2} - 1\}$$

$$P_{wf} = 2823 \text{ psi}$$

According to Vogel equation the reservoir is capable to produce 8000 bpd with $P_{wf} = 2823$ psi.

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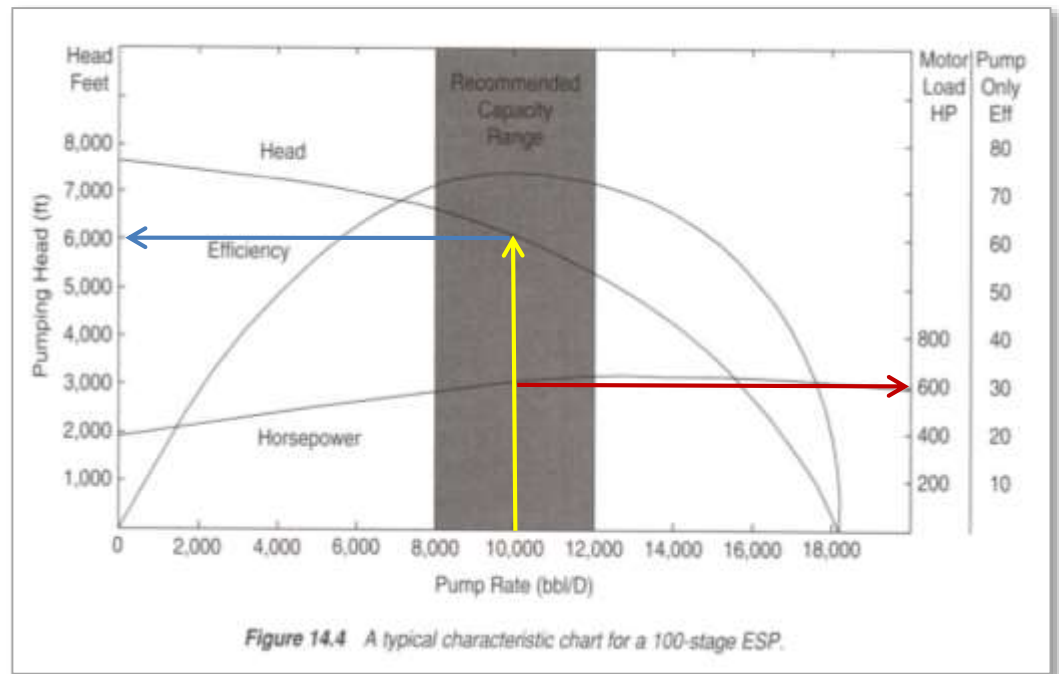
ESP design: Step-by-step procedure

Solution

- Arbitrarily we select an ESP capable to produce a rate 25% higher than the expected production rate. Insofar, the required liquid throughput at pump is:

$$Q_{LD} = (1.25) \times (8.000) = 10.000 \text{ bbl/day}$$

- Select an ESP that delivers liquid flow rate $Q_{LD} = 10.000$ bbl/day in the **neighbourhood of its maximum efficiency.**



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ESP design: Step-by-step procedure

Solution

4. The **minimum pump depth** is:

$$\begin{aligned} D_{\text{pump}} &= D - (P_{\text{wf}} - P_{\text{suction}}) / (0.433 \times SG_L) \\ D_{\text{pump}} &= 10000 - (2823 - 200) / (0.433 \times 0.865) \\ D_{\text{pump}} &= \mathbf{2997 \text{ ft}} \end{aligned}$$

To maximise present and future gas separation efficiency use **pump depth** of: $10000 - 200 = \mathbf{9800 \text{ ft}}$

5. Being pump depth 9800 ft, the **pump suction pressure** is:

$$\begin{aligned} P_{\text{suction}} &= P_{\text{wf}} - (D - D_{\text{pum}}) \times (0.433 \times SG_L) \\ P_{\text{suction}} &= 2823 - (10000 - 9800) \times (0.433 \times 0.865) \\ P_{\text{suction}} &= \mathbf{2748 \text{ psia}} \end{aligned}$$

cont/...

ESP design: Step-by-step procedure

6. Calculate the **outflow pressure**:

$$P_{\text{out}} = P_{\text{THP}} + P_{\text{gravity}} + P_{\text{fr}}$$

a) $P_{\text{THP}} = \mathbf{100 \text{ psi}}$

b) $P_{\text{gravity}} = \text{Fluid Gradient} \times \text{Elevation}$

$$\text{Fluid Gradient} = \text{Fluid density} / 144$$

$$\text{Fluid density} = \text{Fluid SG} \times 62.366 \text{ (water density)}$$

$$\text{Fluid Gradient} = \text{Fluid SG} \times 62.366 / 144 =$$

$$= 0.865 \times (62.366 / 144) = \mathbf{0.865 \times 0.433 = 0.3745 \text{ psi/ft}}$$

$$P_{\text{gravity}} = 0.375 \times 9800 = \mathbf{3671 \text{ psi}}$$

c) $P_{\text{fr}} = F \times G_{\text{avg}}$

where:

F=Friction factor = Unitary friction factor (f) x measured depth

G_{avg} = Average Fluid Gradient

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ESP design: Step-by-step procedure

The unitary friction factor (**f**) for tubing ID 2.992 in at 8000 bpd, according to Hazen-Williams equation is:

$$f = 2.083 \left(\frac{100Q}{34.3C} \right)^{1.852} ID^{-4.8655} = 2.083 \left(\frac{100 \times 8000}{34.3 \times 120} \right)^{1.852} 2.992^{-4.8655} = 174 \text{ ft}/1000 \text{ ft}$$

F = unitary friction factor (f) x measured depth

$$F = 174 \text{ (ft}/1000 \text{ ft)} \times (9.8 \text{ thousand ft}) = \mathbf{1705 \text{ ft}}$$

$$P_{fr} = F \times \text{Fluid Gradient } (G_{avg}) = 1705 \text{ ft} \times 0.375 \text{ psi/ft} = \mathbf{639 \text{ psi}}$$

Therefore :

$$P_{out} = P_{THP} + P_{grvt} + P_{fr} = 100 + 3671 + 638 = \mathbf{4409 \text{ psi}}$$

7. The **required pump pressure differential** is:

$$\Delta P = P_{discharge} - P_{suction} = 4409 - 2748 = \mathbf{1661 \text{ psi}}$$

8. The required **pumping head (H) or Total Dynamic Head (TDH)** is:

$$H = (2.31 \times \Delta P) / SG_L = (2.31 \times 1661) / 0.865 = \mathbf{4436 \text{ ft}}$$

ESP design: Step-by-step procedure

9. At throughput 10.000 bbl/day, the pump characteristic chart gives a pumping head of 6000 ft for the 100-stages pump, which yields 60 ft pumping head per stage.

The required **number of stages** is: $4436/60 = \mathbf{74 \text{ stages}}$

10. At throughput 10.000 bbl/day, the pump characteristic chart gives the power of the 100-stages of 600 hp, which yield 6 hp/stage.

The **required power (W)** for 74 stages is:

$$\mathbf{W = W_{water} \times SG_{produced \text{ fluid}}}$$

$$6 \times 74 \times 0.865 = \mathbf{384 \text{ hp}}$$

ESP design: Step-by-step procedure

Summary of results

1. Minimum pump depth:	2997 ft
2. Pump suction pressure @ 9800 ft:	2748 psi
3. Outflow pressure:	4409 psi
— P_{THP}	= 100 psi
— $P_{gravity}$	= 3671 psi
— P_{fr}	= 638 psi
—Friction factor	= 1705 ft
4. Required pump pressure differential:	1661 psi
5. Pumping head (H) or Total Dynamic Head (TDH):	4436 ft
4. Number of stages:	74
5. Required power :	384 hp60

The same results can be obtained applying the Total Dynamic Head (TDH) approach

ESP design: Step-by-step procedure

Total Dynamic Head (TDH) approach

$$\text{TDH} = 2.31/\text{SG}_{\text{pf}} \times (\text{P}_{\text{THP}} - \text{CHP}) + \text{NLV} + \text{H}_{\text{fr}} \text{ and}$$

$$\text{NLV} = \text{TVD} - \text{P}_{\text{wf}}/\text{G}_{\text{avg}}$$

where:

- **SG_{pf}** is the specific gravity of the produced fluid = 0.865
- **G_{avg}** is the average produced fluid gradient = 0.3745 psi/ft
- **P_{THP}** is the tubing wellhead pressure = 100 psi
- **CHP** is the casing-head pressure, in such case = zero psi
- **P_{wf}** is the bottom-hole flowing pressure = 2823 psi
- **NLV** is the TVD of dynamic fluid level ft
- **TVD** is the Total vertical depth = 10.000 ft
- **H_{fr}** is the tubing frictional head loss, equal to the Friction factor **F** = 1705 ft

$$\text{NLV} = \text{TVD} - \text{P}_{\text{wf}}/\text{G}_{\text{avg}} = 10.000 - 2823/0.3745 = 2462 \text{ ft}$$

$$\begin{aligned} \text{TDH} &= 2.31/\text{SG}_{\text{pf}} \times (\text{P}_{\text{THP}} - \text{CHP}) + \text{NLV} + \text{H}_{\text{fr}} = \\ &2.31/0.865 \times 100 + 2462 + 1705 = 4436 \text{ ft} \end{aligned}$$

Workshop Session - ESP basic design

Workshop session

ESP basic design

Problem

Using the same data of previous example, determine the new required specifications for an ESP supposing to complete the well with a 4½ -in (4.00 in ID) tubing and compare the two cases.

Solution

The calculations from step 1 to 4 are the same.

Workshop session: ESP basic design

5. Calculate the **outflow pressure**:

$$P_{\text{out}} = P_{\text{THP}} + P_{\text{gravity}} + P_{\text{fr}}$$

a) $P_{\text{THP}} = \mathbf{100 \text{ psi}}$ (the same)

b) $P_{\text{gravity}} = \text{Fluid Gradient} \times \text{Elevation}$

Fluid Gradient = Fluid density / 144

Fluid density = Fluid SG x 62.366 (water density)

Fluid Gradient = Fluid SG x 62.366 / 144 =

= 0.865 x (62.366 / 144) = **0.865 x 0.433 = 0.375 psi/ft**

$$P_{\text{gravity}} = 0.375 \times 9800 = \mathbf{3671 \text{ psi}}$$

c) $P_{\text{fr}} = F \times G_{\text{avg}}$

where:

F = Friction factor = Unitary friction factor (f) x measured depth

G_{avg} = Average Fluid Gradient

The P_{fr} will change due to the different TBG ID adopted.

Workshop session: ESP basic design

The unitary friction factor (**f**) for tubing ID 4.00 in at 8000 bpd, according to Hazen-Williams equation is:

$$f = 2.083 \left(\frac{100Q}{34.3C} \right)^{1.852} ID^{-4.8655} = 2.083 \left(\frac{100 \times 8000}{34.3 \times 120} \right)^{1.852} 4.00^{-4.8655} = 44.5 \text{ ft/1000 ft}$$

F = unitary friction factor (f) x measured depth

$$F = 44.5 \text{ (ft/1000 ft)} \times (9.8 \text{ thousand ft}) = \mathbf{436 \text{ ft}}$$

$$P_{fr} = F \times G_{avg} = 436 \text{ ft} \times 0.372 \text{ psi/ft} = \mathbf{162 \text{ psi}}$$

Therefore :

$$P_{out} = P_{THP} + P_{grvt} + P_{fr} = 100 + 3671 + 162 = \mathbf{3933 \text{ psi}}$$

6. The **required pump pressure differential** is:

$$\Delta P = P_{discharge} - P_{suction} = 3933 - 2748 = \mathbf{1185 \text{ psi}}$$

7. The required **pumping head (H)** is:

$$H = (2.31 \times \Delta P) / SG_L = (2.31 \times 1185) / 0.865 = \mathbf{3151 \text{ ft}}$$

Workshop session: ESP basic design

8. At throughput 10000 bbl/day, the pump characteristic chart gives a pumping head of 6000 ft for the 100-stages pump, which yields 60 ft pumping head per stage.

The required **number of stages** is: $3151/60 = \mathbf{53 \text{ stages}}$

9. At throughput 10000 bbl/day, the pump characteristic chart gives the power of the 100-stages of 600 hp, which yield 6 hp/stage.

The **required power (W)** for 53 stages is:

$$\mathbf{W = W_{water} \times SG_{produced \text{ fluid}}}$$

$$6 \times 53 \times 0.865 = \mathbf{275 \text{ hp}}$$

Workshop session

ESP basic design

		case (a)	case (b)	difference	
TBG ID	in	2.992	4.000	(b) - (a)	%
P_{fr}	psi	638	162	-476	-75
P_{out}	psi	4.409	3.933	-476	-11
$P_{dis} - P_{suc}$	psi	1.661	1.185	-476	-29
Head	ft	4.436	3.151	-1285	-29
N_r stages	nr	74	53	-21	-28
Power	Watt	384	275	-109	-28

Recommended Operating Range (ROR)

Operational Factors affecting the ESP performance

ESP Recommended Operating Range - ROR

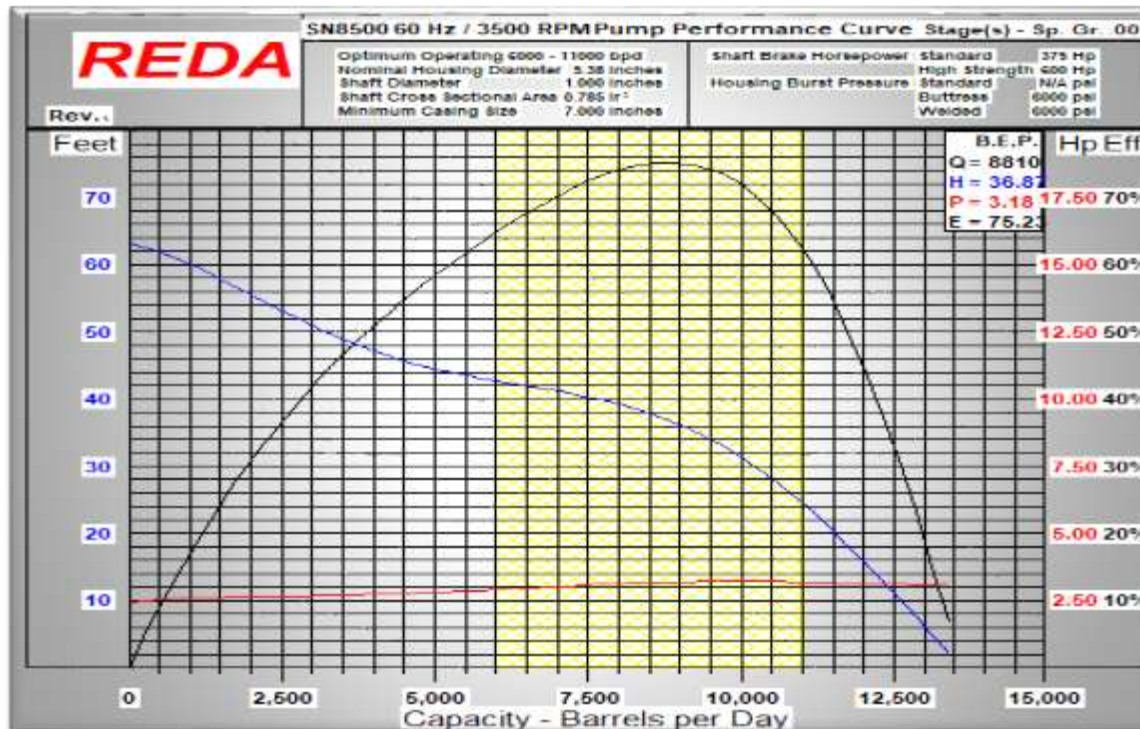
At the end of this section, you will be able to Understand:

- What causes down-thrust and up-thrust in a pump impeller
- The concept of net thrust in a pump impeller
- The difference among Floater and Compression Stage Construction
- The significance of the recommended operating range for a given pump.
- The Cavitation phenomenon and the way to avoid or reduce this damaging event

ESP Recommended Operating Range - ROR

A good guideline for appropriate volumes for a given pump stage is the **recommended operating range (ROR)**. This is highlighted in yellow on the pump curve.

This range is based on a combination of factors, such as the relative efficiency and the efficiencies of other pumps...



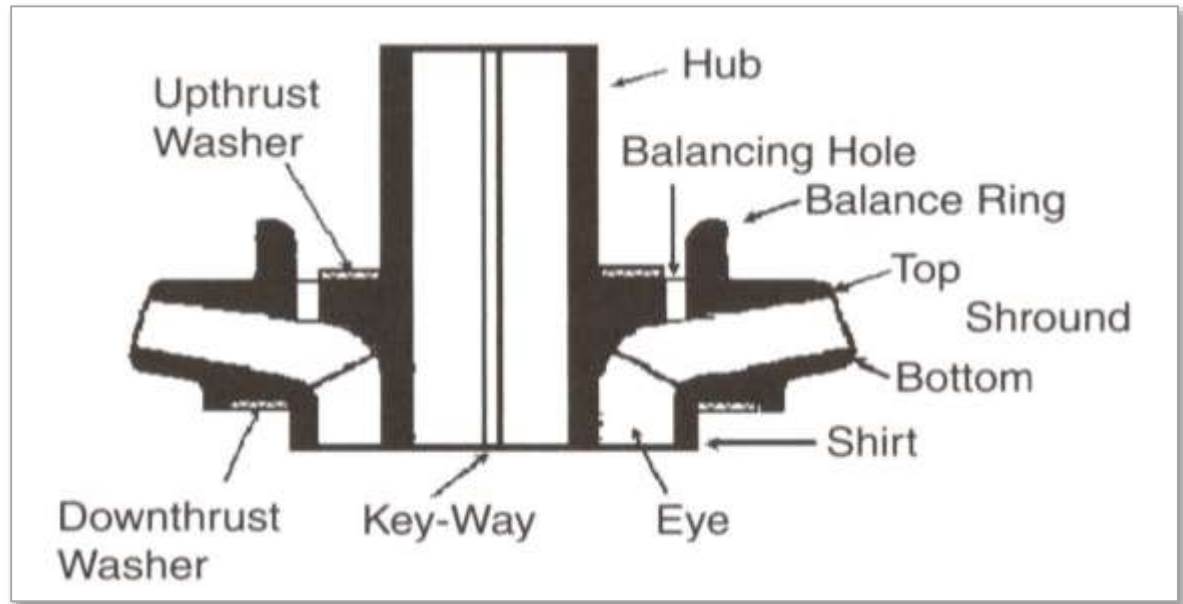
...but mainly the criteria is acceptable pump run-life due to THRUST and CAVITATION

Axial Thrust Forces

ESP Recommended Operating Range - ROR

Axial Thrust Forces

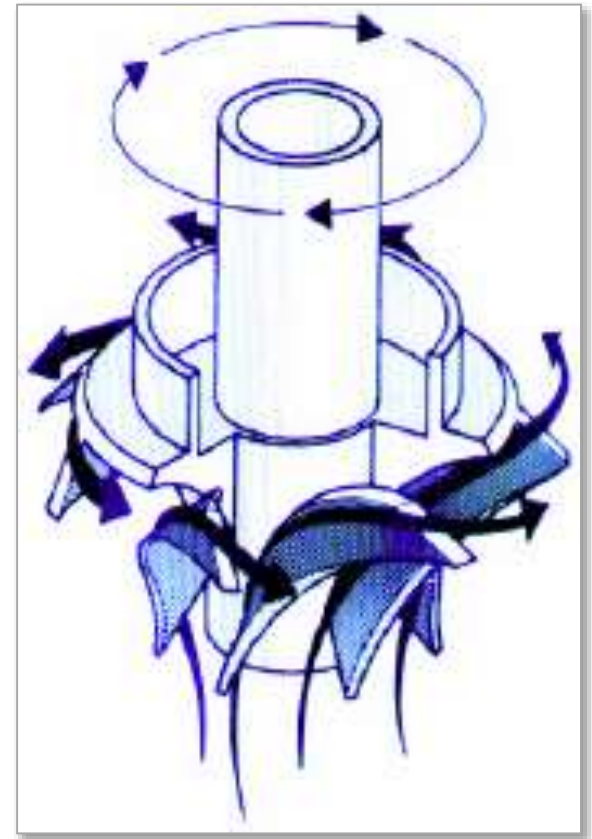
- During the pumping action, various **unbalanced forces** arise on the centrifugal pump's impeller and these forces are directly transmitted to the pump shaft.
- The **radial components** of these forces are taken up by the housing of the ESP pump and do **not significantly affect** the proper operation.
- **Axial force** components **are much more detrimental** if not taken up by thrust bearings.



ESP Recommended Operating Range - ROR

Axial Thrust Forces

- Axial forces in ESP pump can be classified in two groups:
 - Static and
 - Dynamic
- **Static forces** arise due to the weight of pump parts in the produced fluid and act downward:
 - the weight of the impellers, and
 - the weight of the pump shaft
- **Dynamic forces** are the results of pumping action and are related to the flow of the produced fluid through the pump's stages.

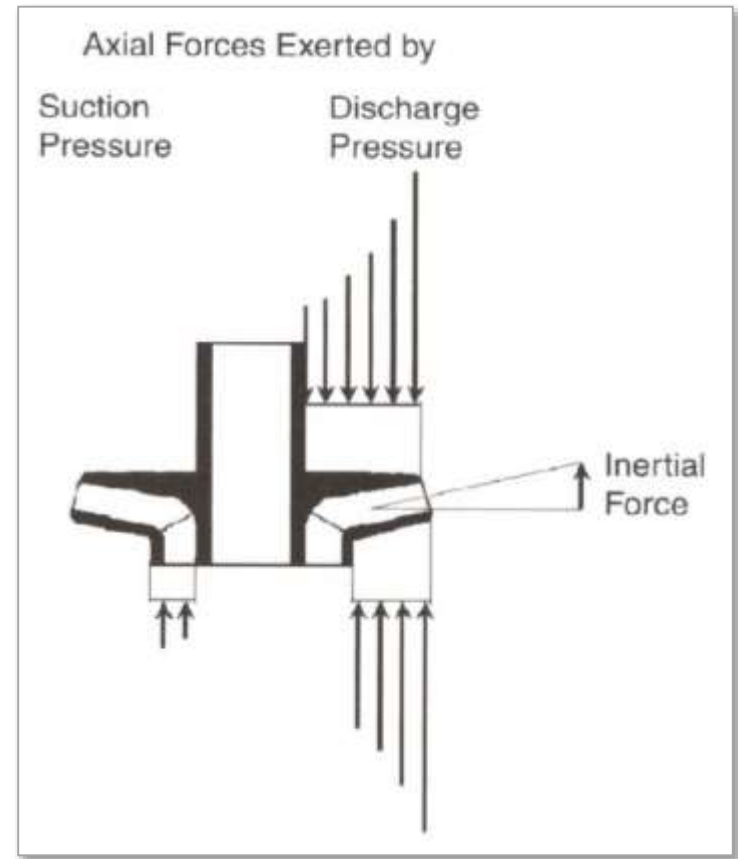


ESP Recommended Operating Range - ROR

Dynamic forces

The Dynamic forces take the following form:

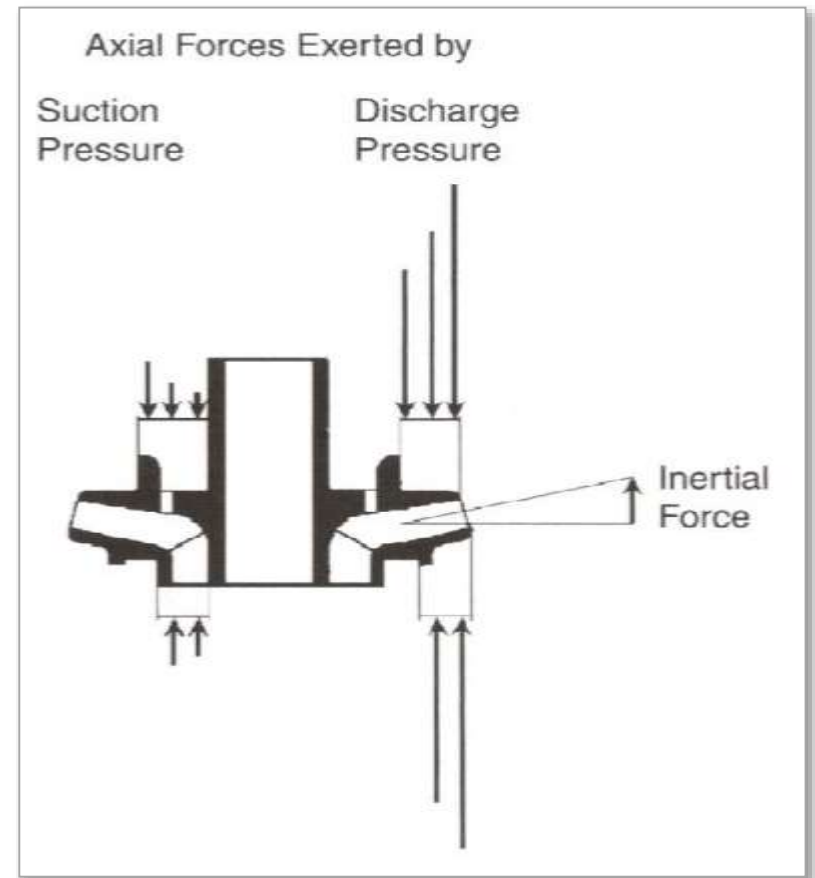
- Forces resulting from the suction and discharge pressures acting on the two shrouds of the impeller. **The net of these forces always acts downward.**
- The **net inertial force (momentum)** due to the change in flow direction inside the pump stage. Since the velocity of the produced fluid inside the impeller is much higher at the discharge side than at the suction, **this force always acts upward.**
- The axial load due to the pump discharge pressure acting on the cross-section area of the pump shaft, **acting downward.**



ESP Recommended Operating Range - ROR

Dynamic forces

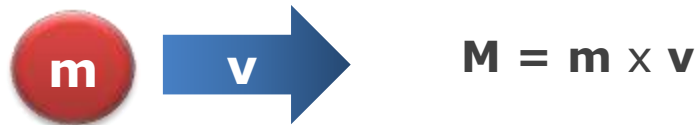
- The forces are greater toward the periphery of the impeller because of the rotation of the fluid.
- A balancing rig and balancing holes drilled in the top shroud of the impeller are used as to reduce the axial thrust.
- These modifications reduce the volumetric efficiency of the stage but at the same time greatly reduce the axial forces acting on the impeller.



ESP Recommended Operating Range - ROR

Momentum

As everybody know the **momentum (M) of a body is defined as the product of its mass (m) and velocity (v)**


$$\mathbf{M} = \mathbf{m} \times \mathbf{v}$$

Newton's second law

Newton's second law states that the force applied to a body produces a proportional acceleration; the relationship between the two is: **$\mathbf{F} = \mathbf{ma}$**

where:

F is the force applied, **m** is the mass of the body, and **a** is the body's acceleration.

If the body is subject to **multiple forces** at the same time, then the acceleration is proportional to the *vector sum* (that is, the *net force*): **$\mathbf{F}_1 + \mathbf{F}_2 + \dots + \mathbf{F}_n = \mathbf{F}_{\text{net}} = \mathbf{ma}$**

The second law can also be shown to relate the **net force** and the **momentum M** of the body:

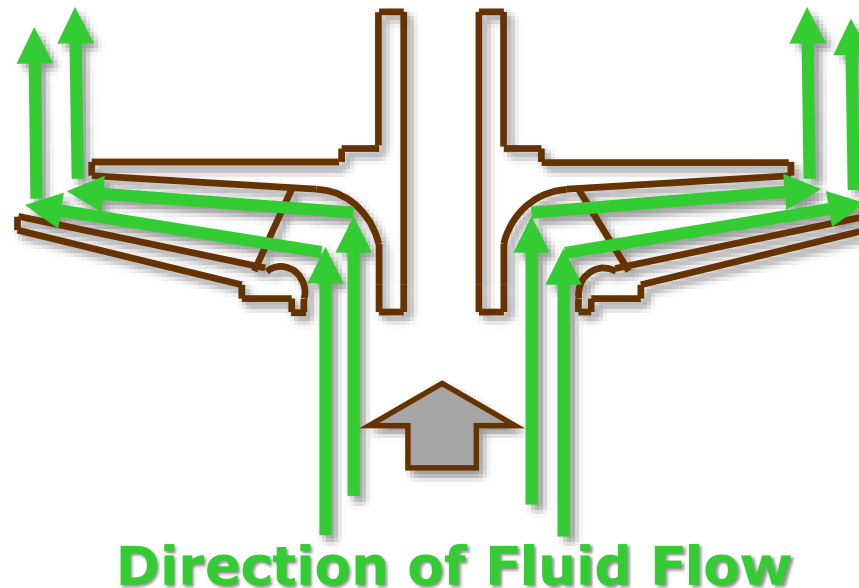
$$\mathbf{F}_{\text{net}} = \mathbf{ma} = \mathbf{m} \left(\frac{d\mathbf{v}}{dt} \right) = \frac{d(\mathbf{mv})}{dt} = \frac{d\mathbf{M}}{dt}$$

The variation of the momentum (due to velocity variation) generates a net force.

ESP Recommended Operating Range - ROR

Inertial Forces

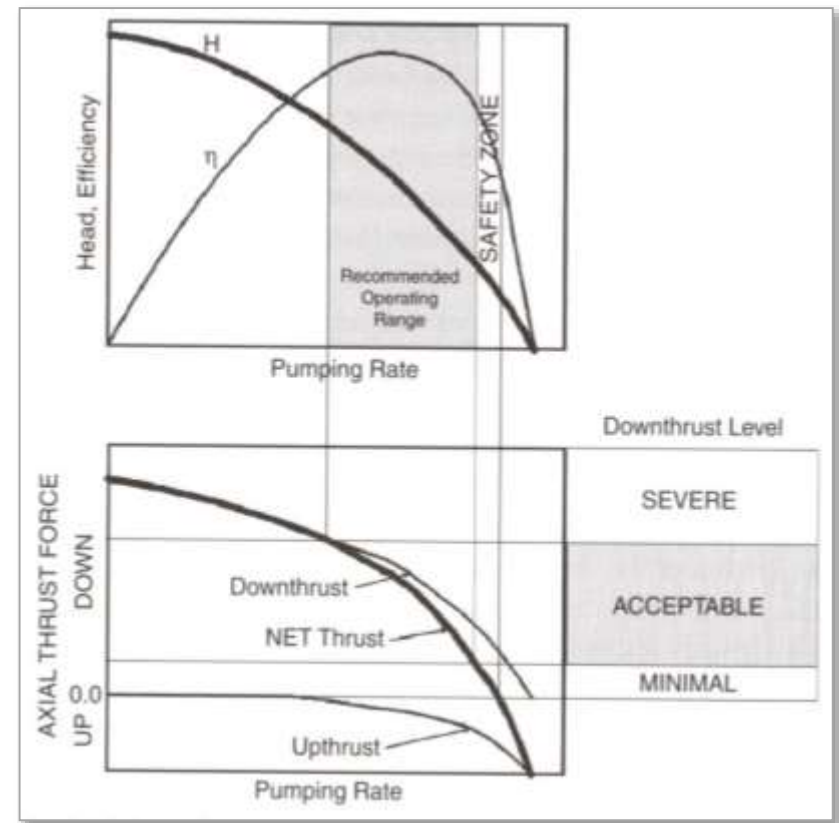
- As already mentioned, the fluid entering the bottom of the impeller is forced to change direction. This *change in momentum exerts an upward force on the impeller*.
- This is the ***only one that creates up-thrust***



ESP Recommended Operating Range - ROR

Axial thrust versus pumping rate

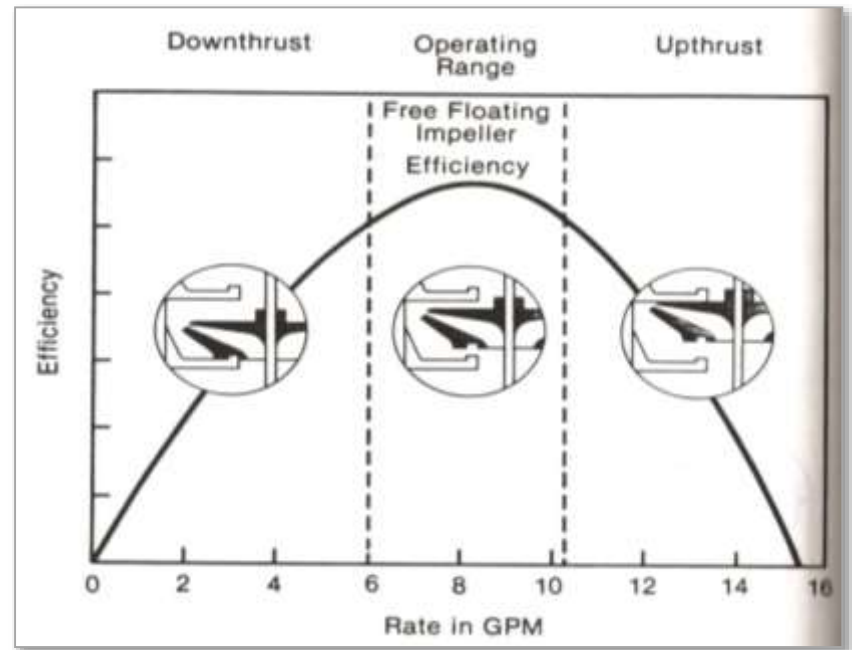
- Down-thrust is basically determined by the head developed since its main component comes from the pump's discharge pressure acting on the top and bottom shrouds of the impeller.
- Its variation with pumping rate, therefore, follows the shape of the pump's head-rate performance curve.
- It is at a maximum at shut-in conditions: pump running with discharge closed.
- Up-thrust forces are the result of the change in inertial forces and are proportional to the kinetic energy of the liquid pumped. Thus their variation with pumping rate follows a second-order curve.
- The sum of up- and down-thrust gives the net thrust.



ESP Recommended Operating Range - ROR

Net Thrust

- The **net thrust** on an impeller is the sum of three components:
 1. Pressure causing **down-thrust**
 2. The **weight** of the impeller causing **down-thrust**
 3. Change in fluid momentum causing **up-thrust**.
- The impellers are designed to **be balanced near the best efficiency point (BEP)** of the stage.
- To ensure ideal conditions ESP pumps must be operated at the rate belonging to that point.
- Since in real applications this point is almost impossible to accomplish, there is always an axial thrust acting on the pump shaft that must be taken care of.

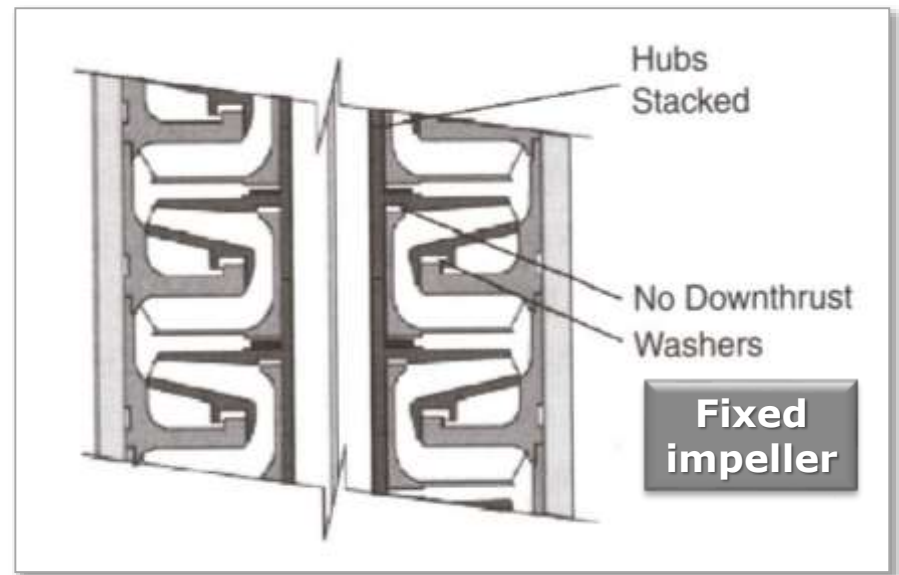
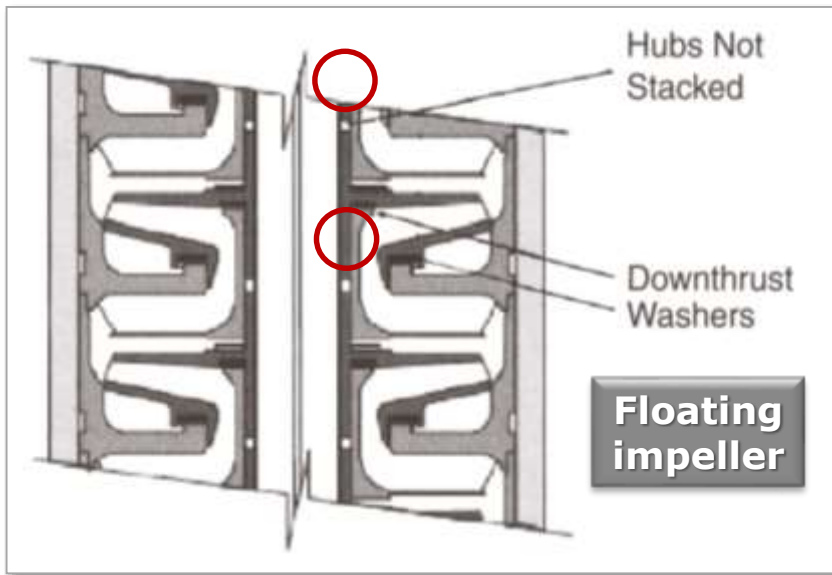


ESP Recommended Operating Range - ROR

Axial forces compensation

The axial forces developed in ESP pump must be compensated, otherwise the axial movement of the impellers and the pump shaft lead to mechanical damage of the stages.

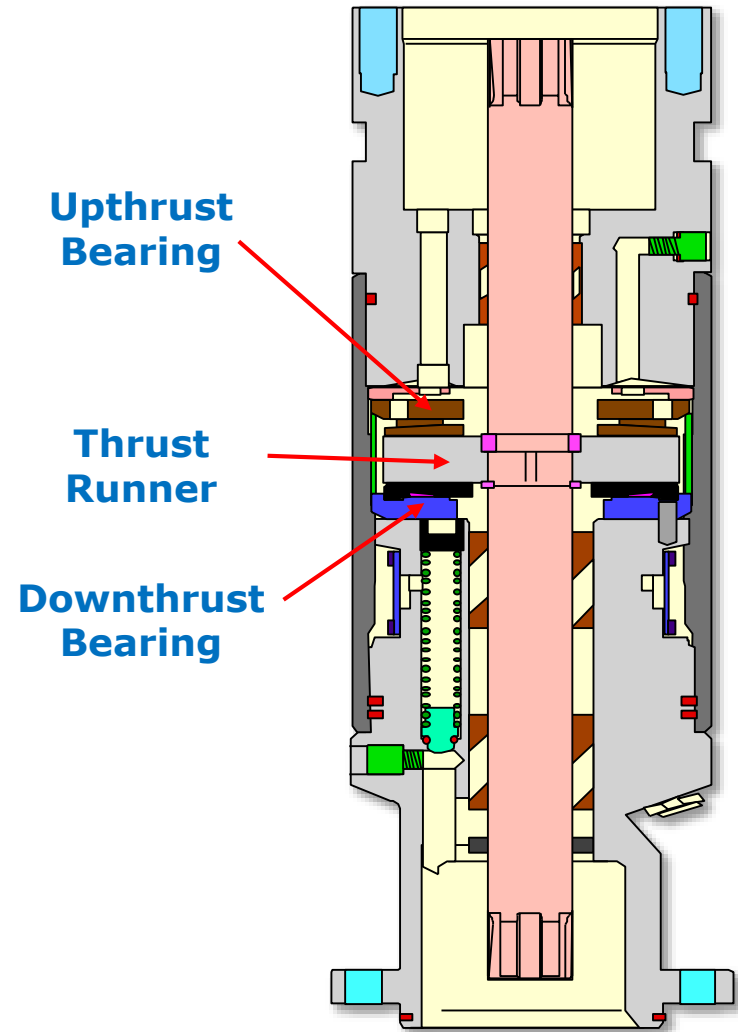
Elimination of such forces is accomplished differently in stages with **fixed impellers** from stages with **floating impellers**.



ESP Recommended Operating Range - ROR

Fixed vs Floating impellers

- In **fixed** impeller pumps all axial forces **are transmitted to the pump shaft** and have to be balanced by the **main thrust bearing, situated in the protector section** of the ESP unit. This solution necessitates the use of thrust bearing of relatively large capacity.
- In **floating** impeller pumps most of the **axial forces are compensated by frictional forces** arising in the up- and down-thrust washers installed on the impellers, requiring smaller-capacity thrust bearing.



ESP Recommended Operating Range - ROR

Floating impellers

The **benefits** of floating impeller design include:

- The elimination of having to fix the impellers axially, a time-consuming work requiring high precision.
- The building of pumps with several hundreds of stage is possible.
- Smaller capacity thrust bearings are needed in the protector section because most of the hydraulic thrust is absorbed inside the pump.
- Lower investment cost, as compared to fixed impeller pumps.

Limitations are related to the load bearing capacity of available thrust bearings, which, in turn, are restricted by the annular space available:

- Such pump are usually manufactured in **smaller diameters**, up to a size of about 6 in., and
- The recommended **operating range is somewhat narrower** than for the same pump with fixed impeller.

ESP Recommended Operating Range - ROR

Fixed impellers

- Since the stages are not equipped with down-thrust washers, the axial thrust developed on them must be fully carried by the unit's main thrust bearing in the protector section.
- Pumps with such stages are often called "compression" pumps and are commonly used in large-size ESP units (diameter greater than 6 in.)

Benefits of fixed impeller design include:

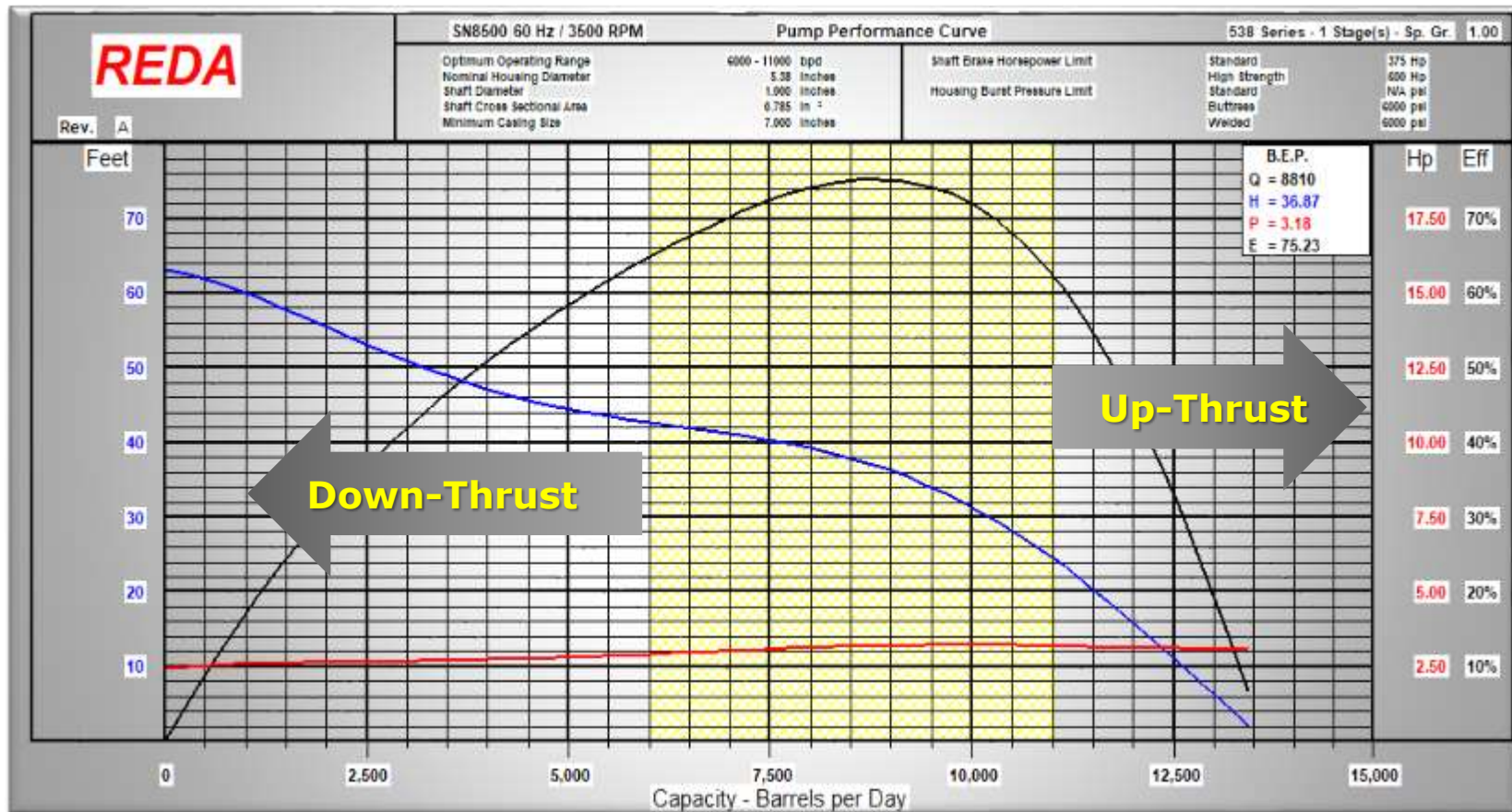
- They are capable to produce large volumes of liquid.
- They may have a wider operating range than pumps of the same type with floating impellers.

Limitations of fixed impeller design include:

- They are most difficult to manufacture because impellers must be fitted very precisely along the pump shaft.
- Higher investment cost.
- The maximum number of stages in one pump is limited to about 80 to 100.
- Protector with high capacity thrust bearings must be used.

ESP Recommended Operating Range - ROR

Based on the previous discussion, there is some logic to labeling the curve as shown below.



ESP Recommended Operating Range - ROR

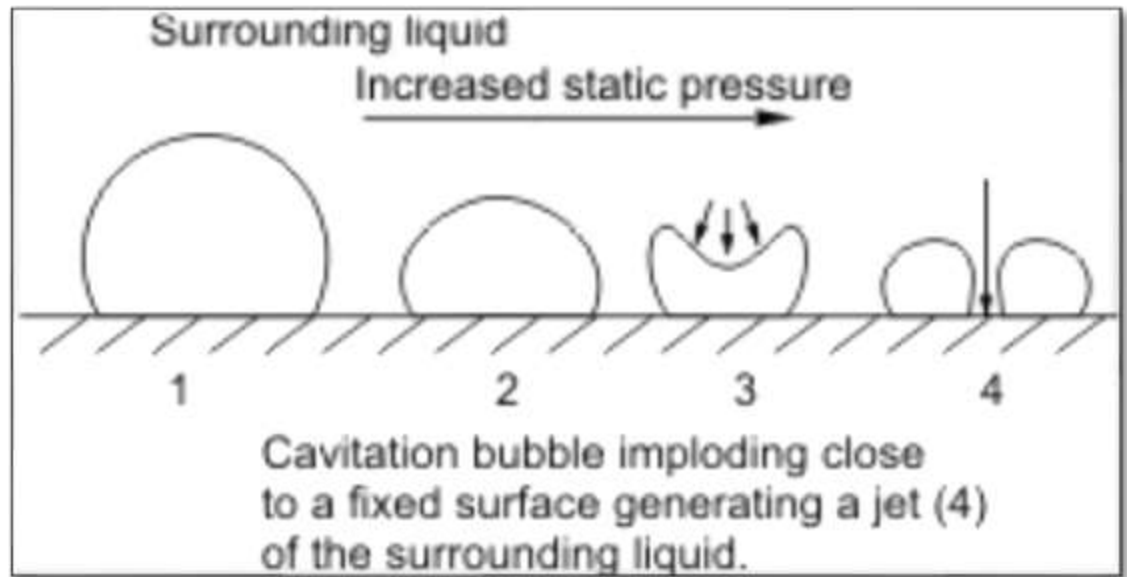
Cavitation

Cavitation may occur when the pressure at any place inside the pump **fall below the saturated vapor pressure** of the liquid.

This involves the **formation of small vapor bubbles** which, when taken by the flowing liquid to places at higher pressure, will suddenly **collapse producing a shock wave**.

Although the collapse of a cavity is a relatively low-energy event, **highly localized collapses can erode metals**, such as steel, over time.

The pitting caused by the collapse of cavities produces great wear on components and **can dramatically shorten a pump's lifetime**.



ESP Recommended Operating Range - ROR

Cavitation

Cavitation may occur in the ESP pump impeller's eye where a **great increase of velocity** take place.

This, according to Bernoulli's law, involves a **sudden decrease of flowing pressure**.

Cavitation can be prevented by the presence of a adequate length of liquid column (pressure) above the pump intake.

ESP Recommended Operating Range - ROR

Some examples of impeller damaged by cavitation



Electricity and Magnetism - Review of Fundamentals

Electricity and Magnetism: Review of Fundamentals

At the end of this section, you will be able to understand the Electrical fundamentals relating to:

- Alternating current
- AC circuit
- AC power

Electricity and Magnetism: Review of Fundamentals

Three-phase electric power

Three-phase electric power is a common method of **alternating-current** electric power transmission.

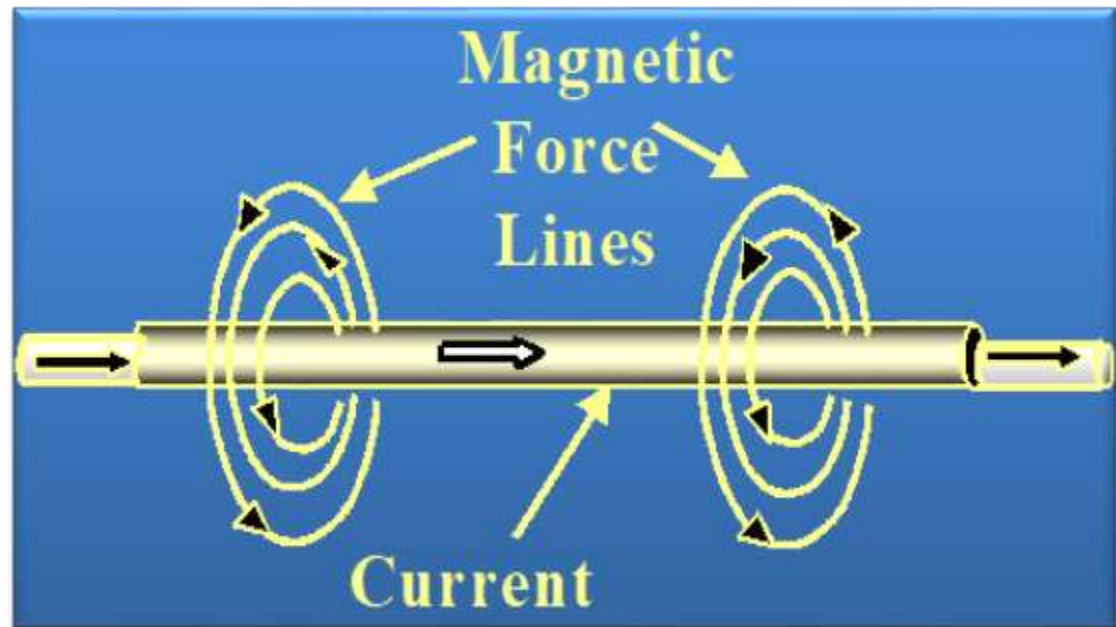
It is a type of **polyphase system**, and is the most common method used by electric power distribution grids worldwide to distribute power.

It is also used **to power large motors** and other large load.

Electricity and Magnetism: Review of Fundamentals

Induced Magnetic Field

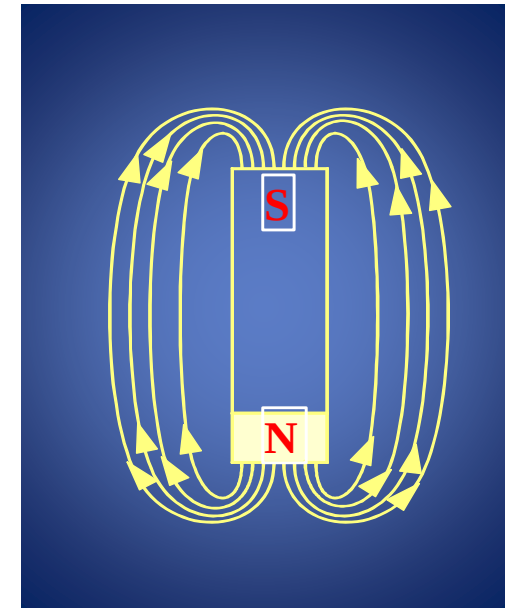
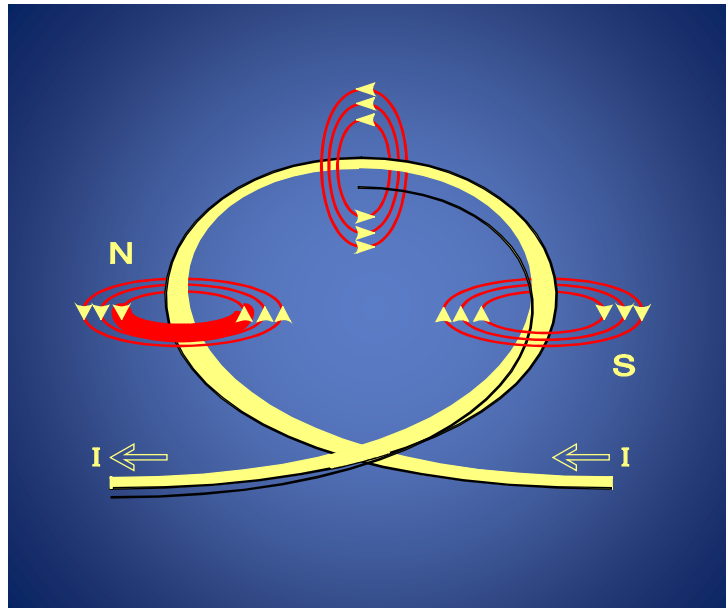
Current flowing in a conductor induces a magnetic field around the conductor.



Electricity and Magnetism: Review of Fundamentals

Electricity and Magnetism

When the wire is looped, the force field created looks very much like a bar magnet has been placed inside it based on the flux lines.



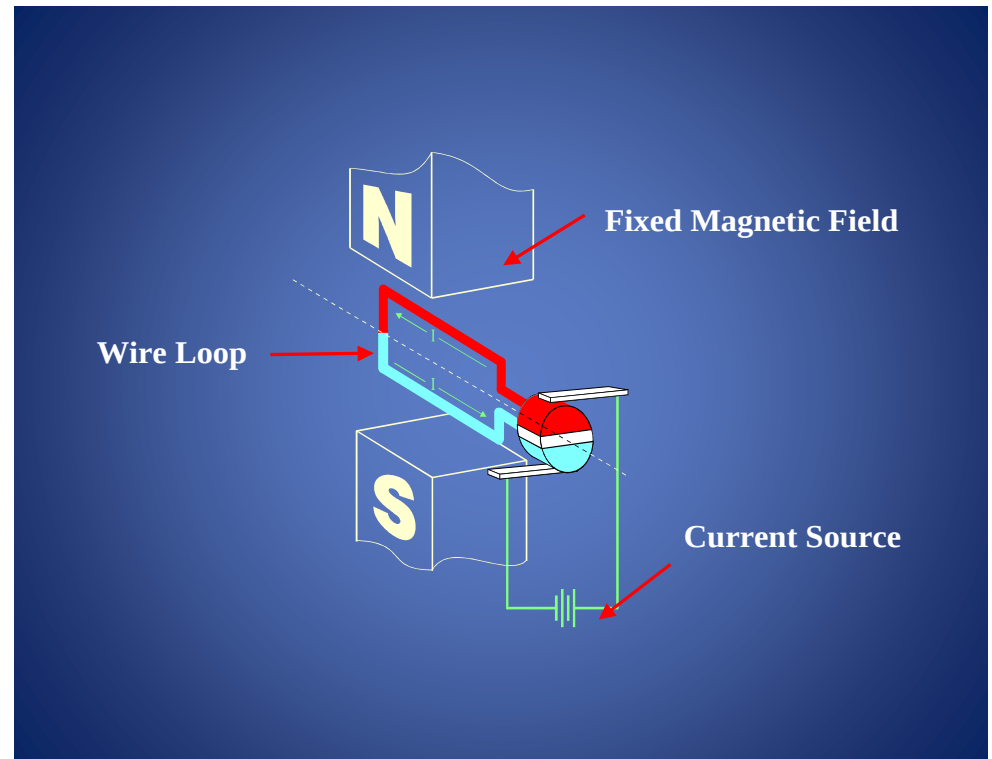
Electricity and Magnetism: Review of Fundamentals

Magnetic Force

If **another wire loop** is placed inside a magnetic field, **nothing will happen to it.**

On the other hand, if **current is flowing** through that wire loop, **it will create a magnetic field around it.**

With **two magnetic fields**, there are **attractive** and **repulsive forces** and thus, a force on the wire loop.

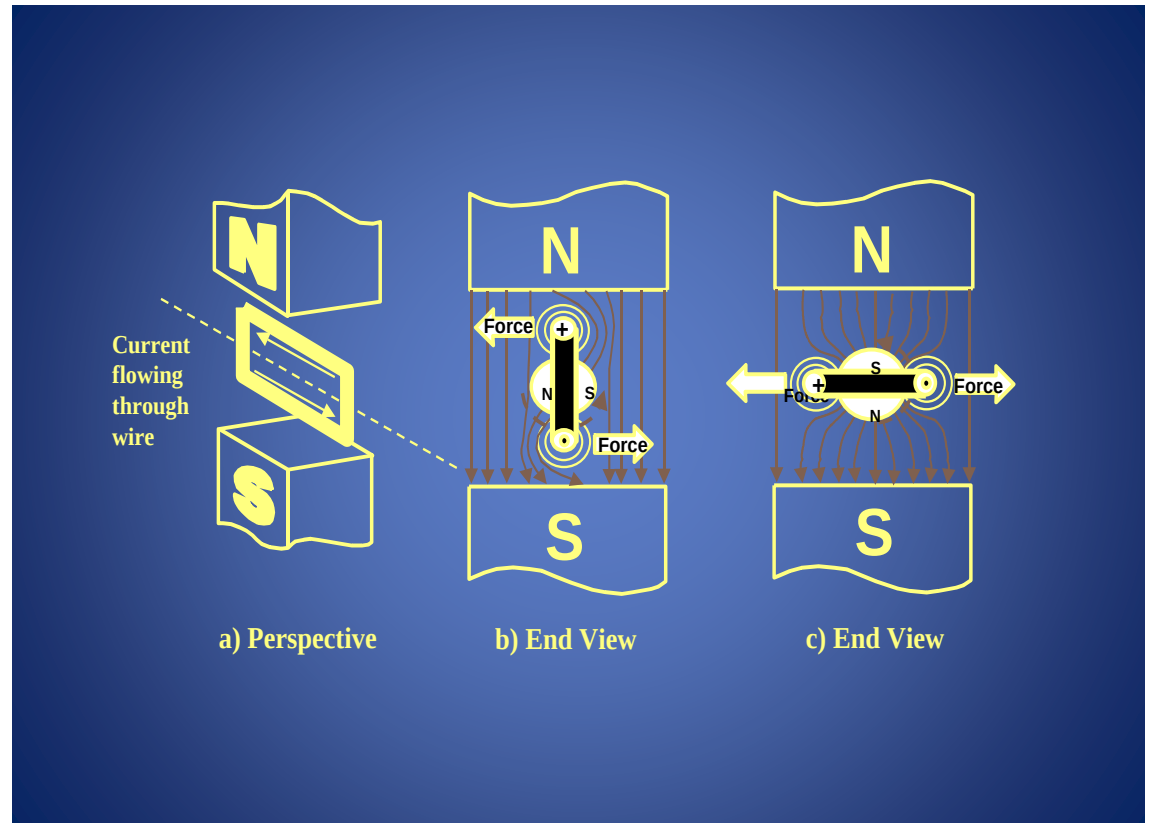


Electricity and Magnetism: Review of Fundamentals

Magnetic Field

If the loop is in line with the magnetic field, the secondary magnetic field will be perpendicular to the main field.

This *will cause two equal and opposite forces* (a torque) on the loop *causing it to rotate until the forces balance*.



Electricity and Magnetism: Review of Fundamentals

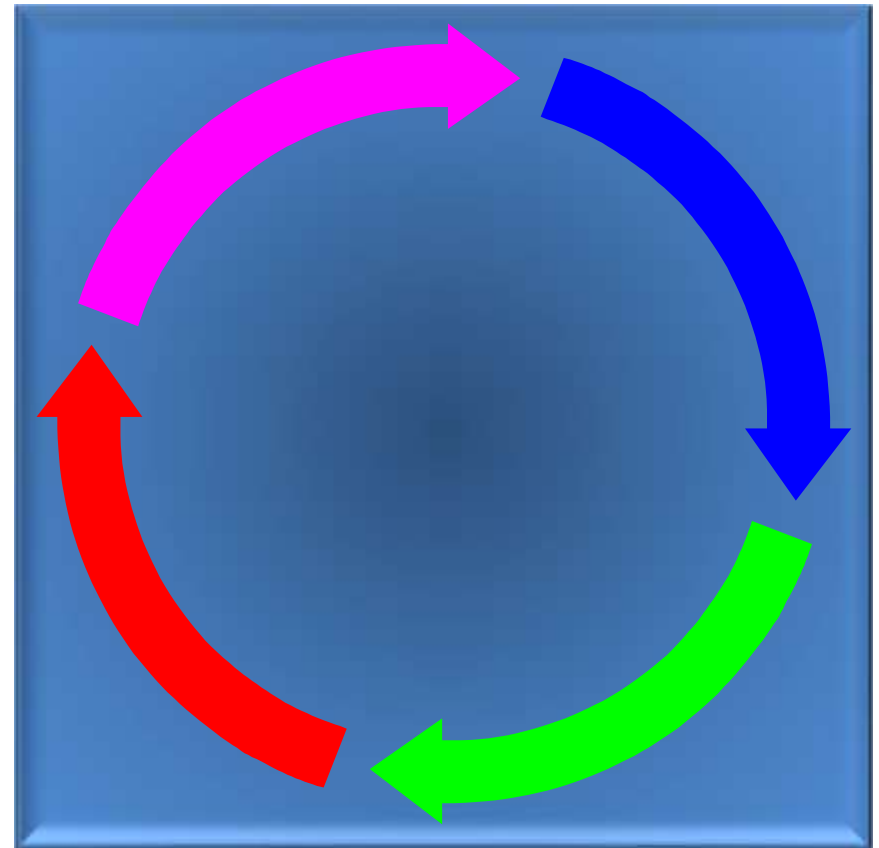
Rotating Magnetic Field

The forces will reach a **steady state** and **hold the magnet in place** as long as current is applied.

This is **not much rotation**.

To cause rotation, **the field must rotate**.

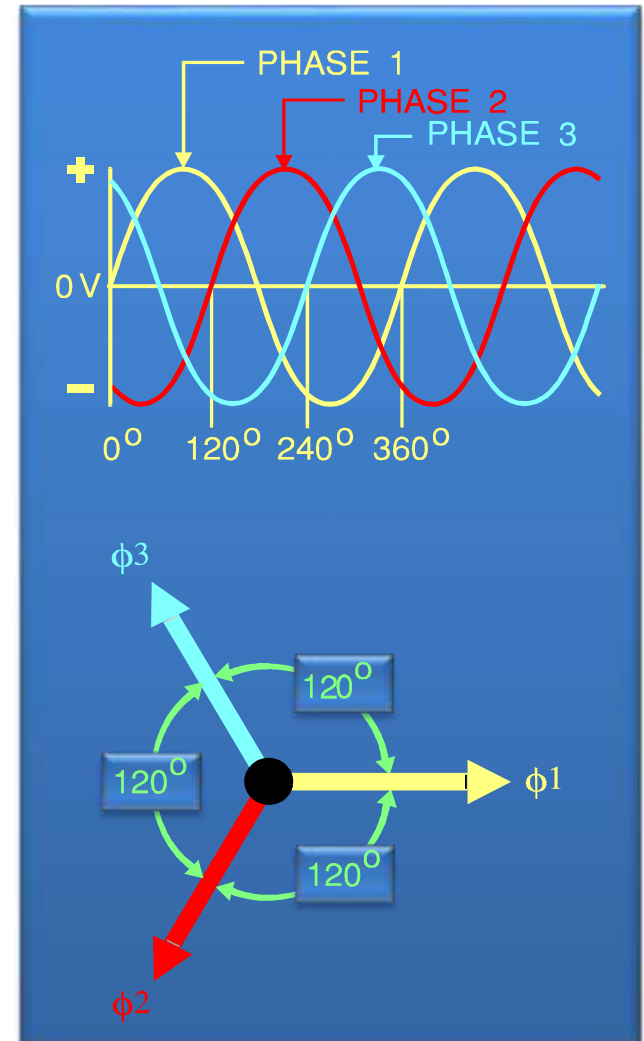
This is accomplished with the alternating current.



Electricity and Magnetism: Review of Fundamentals

This figure illustrates one voltage cycle of a three-phase system, labeled 0 to 360° (2π radians) along the time axis.

The plotted line represents the **variation of instantaneous voltage (or current) with respect to time**. This cycle will repeat **50 or 60 times per second**, depending on the power system frequency.

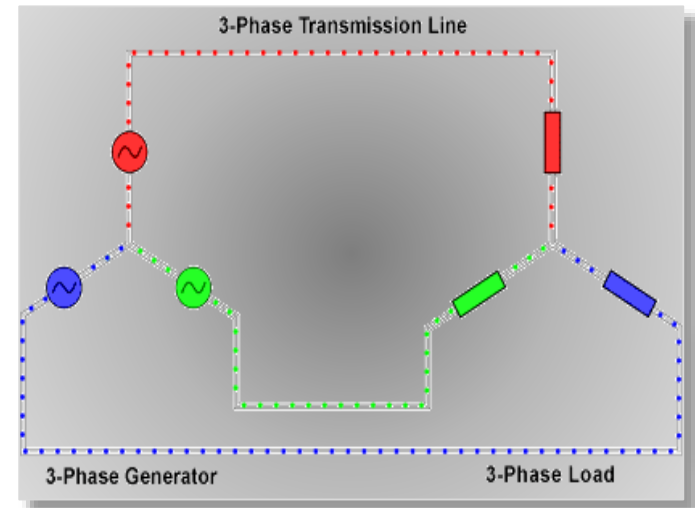


Electricity and Magnetism: Review of Fundamentals

Three-phase electric power

In a three-phase system, **three circuit** conductors carry three alternating currents (of the same frequency) **which reach their instantaneous peak values at different times.**

Taking one conductor as the reference, the other two currents **are delayed in time by one-third and two-thirds** of one cycle of the electrical current.



Three-phase electric motor

The rotating magnetic field of a three-phase motor generated by a three-phase electric power



Electricity and Magnetism: Review of Fundamentals

Motor's rotational Synchronous speed

The **speed of the AC motor** is determined primarily by the **frequency** of the AC supply and the **number of poles** in the stator winding, according to the relation:

$$N_{synch} = \frac{120 f}{p}$$

where:

N_{synch} = Synchronous speed, RPM

f = AC power frequency, Hz

p = Number of poles in the stator

Electricity and Magnetism: Review of Fundamentals

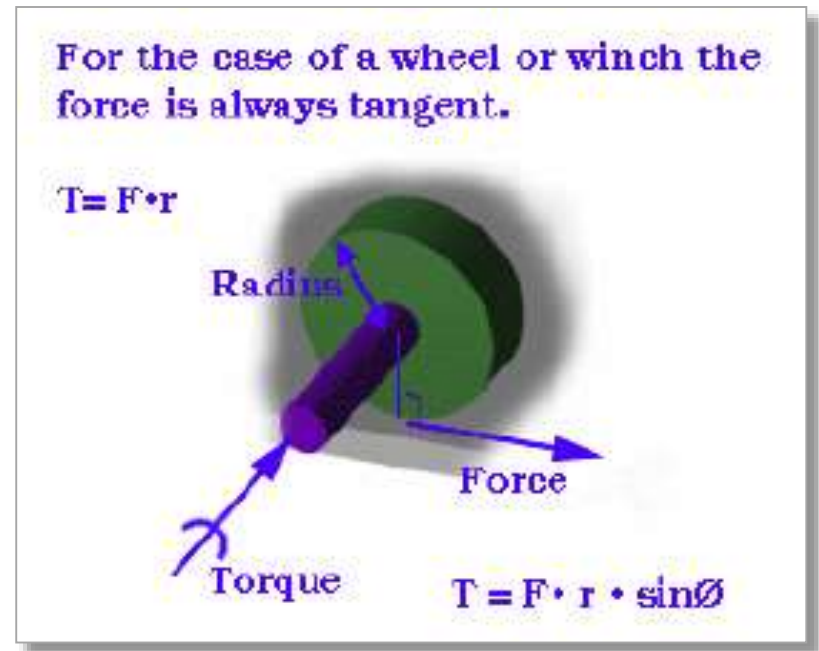
Synchronous speed and Torque

Synchronous speed is the absolute upper limit of motor speed.

At synchronous speed (or **NO load speed**), there is no difference between **rotor speed and rotating field speed**, so no voltage is induced in the rotor bars, **hence no torque is developed.**

Torque is the force that causes an object to rotate (wheels of the car, ESP's shaft). Torque consist of a force acting on distance.

Torque, like work, is measured is pound-feet (lb-ft). However, torque, unlike work, **may exist even though no movement occurs** (you are drive your electrical car with break activated).



Electricity and Magnetism: Review of Fundamentals

Slip and Torque

If the motor is running **under load** (driving the ESP's shaft) the rotor speed will be less than this calculated synchronous speed, in other words, **the rotor rotates slower than the magnetic field.**

So, motor slip is necessary for torque generation.

The rotor speed ***is just slow enough to cause the proper amount of rotor current to flow, so that the resulting torque sufficient to overcome friction losses and drive the load.***

This **speed difference** between the rotor and magnetic field, called **slip**, is normally referred to as a percentage of synchronous speed:

$$S = \frac{N_s - N}{N_s} \times 100$$

where:

S is the slip, %

N_s is the synchronous speed, RPM

N is the actual speed, RPM

Electricity and Magnetism: Review of Fundamentals

Motor Slip

The motor slip **depends on motor parameters**. As seen in the table, smaller motors and lower-speed motors typically have higher relative slip. However, high-slip large motors and low-slip small motors are also available.

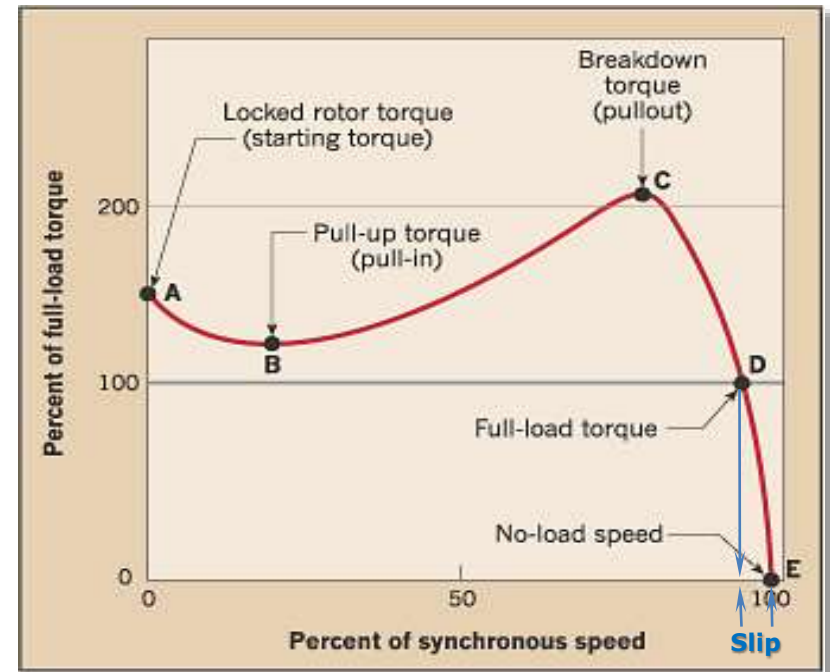
Aluminum				Cast Iron			
NEMA frame	Output (hp)	Speed (Full load RPM)	Slip (%)	NEMA frame	Output (hp)	Speed (Full load RPM)	Slip (%)
48	0.5	3,435	4.6	284T	15	1,170	2.5
48	0.5	1,685	6.4	284T	15	870	3.3
143T	0.5	1,150	4.2				
143T	0.5	845	6.1	326TS	50	3,535	1.8
				326TS	50	1,770	1.7
184T	5	3,505	2.6	365T	50	1,180	1.7
184T	5	1,740	3.3	404T	50	885	1.7
215T	5	1,160	3.3				
254T	5	875	2.8	505TS	250	3,580	0.6
				505TS	250	1,787	0.7
254T	15	3,515	2.4	585	250	1,192	0.7
254T	15	1,760	2.2	586	250	892	0.9

Electricity and Magnetism: Review of Fundamentals

Polyphase motors

National Electrical Manufacturers Association (NEMA) classifies polyphase induction motors according to ***locked rotor torque and current, breakdown torque, pull up torque, and percent slip.***

- **Locked rotor torque**, also called **starting torque** is the minimum torque that the motor develops when it is initially turned on. Starting torque is the amount required to overcome the inertia of a standstill.
- **Pull up torque** is the **minimum torque** developed *during the period of acceleration* from rest to the speed that breakdown torque occurs.
- **Breakdown torque** is the **maximum torque** that the motor develops at rated voltage and frequency, without an abrupt drop in speed. High breakdown torque is necessary for applications that may undergo frequent overloading.
- **Full load torque** is produced by a motor functioning at a rated speed and horsepower. ***The operating life is significantly diminished in motors continually run at levels exceeding full load torque.***
- **Synchronous speed** is the speed at which **no torque** is generated by a motor. ***This occurs in motors that run while not connected to a load.*** At synchronous speed, the rotor turns at exactly the same rate as the stator's rotating magnetic field. Since there is no slip, there is no torque produced.



Electricity and Magnetism: Review of Fundamentals

Polyphase motors

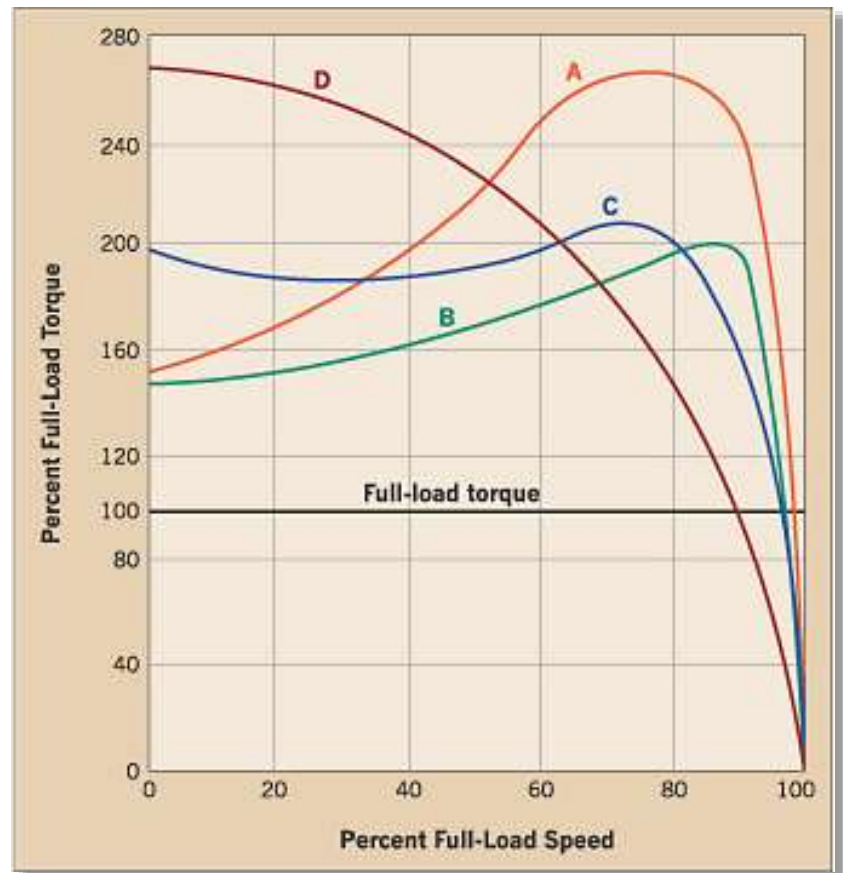
The figure illustrates typical speed-torque curves for NEMA Design A, B, C, and D motors.

Design A motors have a higher breakdown torque than Design B motors and are usually designed for a specific use. Slip is 5%, or less.

Design B motors account for most of the induction motors sold. Often referred to as general purpose motors, slip is 5% or less.

Design C motors have high starting torque with normal starting current and low slip. This design is normally used where breakaway loads are high at starting, but normally run at rated full load, and are not subject to high overload demands after running speed has been reached. Slip is 5% or less.

Design D motors exhibit high slip (5 to 13%), very high starting torque, low starting current, and low full load speed. Because of high slip, speed can drop when fluctuating loads are encountered. This design is subdivided into several groups that vary according to slip or the shape of the speed-torque curve. These motors are usually available only on a special order basis.



Electricity and Magnetism: Review of Fundamentals

Motor's nameplate data

Motors are designed to yield optimal performance when operating at or near by the **nameplate** value.

The nameplate data includes:

- ***Rated voltage***
- ***Rated full-load amps***
- ***Frequency***
- ***Phase***
- ***Rated full-load speed***

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Calculating Full-load Torque

Full-load torque is the torque to produce the rated power at full speed of the motor.

To calculate motor full-load torque, apply this formula:

$$T = \frac{HP \times 5252}{RPM}$$

where:

T = torque, lb-ft

HP = horsepower hp (1hp = 33000 lb-ft/minute)

5252 = constant (33000 divided by $3.14 \times 2 = 5252$)

RPM = revolutions per minute

Example:

What is the FLT (Full-load torque) of a 30HP motor operating at 1725 RPM ?

$$T = \frac{30 \times 5252}{1725} = 91.34 \text{ lb - ft}$$

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Calculating Horsepower

Electrical power is rated in **horsepower or watts**. A horsepower is a unit of power equal to **746 watts or 33,000 lb-ft per minute (550 lb-ft per second)**. A watt is a unit of measure equal to the power produced by a current of 1 amp across the potential difference of 1 volt. It is 1/746 of 1 horsepower. The watt is the base unit of electrical power. Motor power is rated in horsepower and watts.

To calculate the horsepower of a motor when current and efficiency, and voltage are known, apply this formula:

$$HP = \frac{V \times I \times Eff}{746}$$

where:

HP = horsepower, hp

V = voltage, Volt

I = current, Amp

Example:

What is the horsepower of a 230v motor pulling 4 amps and having 82% efficiency ?

$$HP = \frac{230 \times 4 \times 0.82}{746} = \frac{754.4}{746} = 1 \text{ Hp}$$

Electricity and Magnetism: Review of Fundamentals

Calculating Horsepower

To calculate the horsepower of a motor when the speed and torque are known, apply this formula:

$$HP = \frac{RPM \times T}{5252}$$

Example:

What is the horsepower of a 1725 RPM motor with a FLT 3.1 lb-ft ?

$$HP = \frac{1725 \times 3.1}{5252} = \frac{5347.5}{5252} = 1 \text{ Hp}$$

Electricity and Magnetism: Review of Fundamentals

Voltage drop in a Resistance electric circuit

- If an AC current flows through a **resistance component** (incandescence lamp, resistance oven, etc), the voltage drop across it is in the direction of the current, thus **voltage and current are in phase**.
- The magnitude of the **voltage drop** is determined from the well-known Ohm's law:

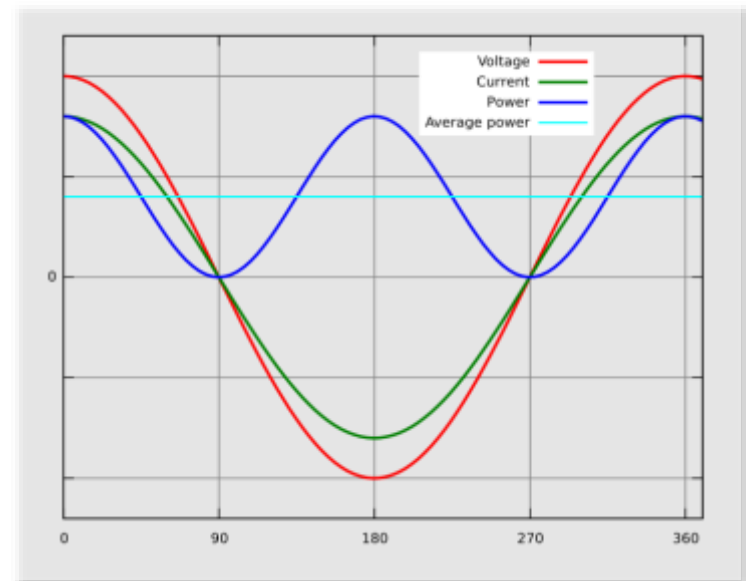
$$U = R \times I$$

where:

U = voltage, V

R = resistance, ohms

I = current, A



Electricity and Magnetism: Review of Fundamentals

Voltage drop in circuit containing resistive, inductive and capacitive components

- In case of **inductances** (like the coils in an electric motor), the flow of electric AC current produces magnetic effect. If the current increases, the system stores energy in the magnetic field which is released when the current starts to decrease. Thus magnetic effect react upon the original current with the end result that **current and voltage are not in phase**.
- The magnitude of the **voltage drop** in such case is determined from the **generalized Ohm's law**:

$$U = Z \times I$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

where:

U = voltage, V

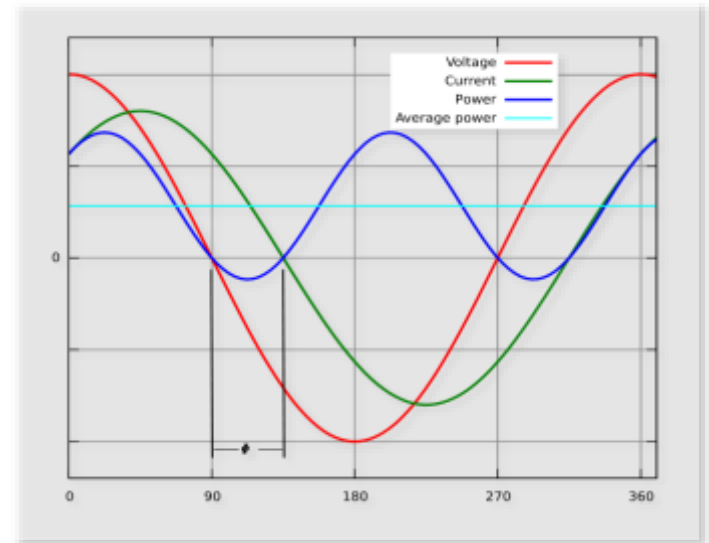
I = current, A

Z = impedance, ohms

R = resistance, ohms

X_L = inductive resistance, ohms

V_C = capacitive resistance, ohms

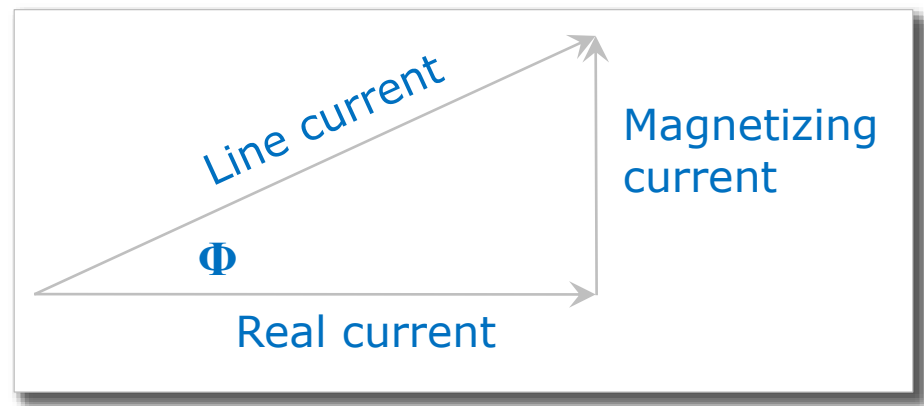


Electricity and Magnetism: Review of Fundamentals

Line Current versus Real current

For an three phase electric AC current induction motor (ESP electric motor) we have an:

- **Line current** and,
- **Real current**



$$I_{\text{real}} = I_{\text{line}} \cos \Phi$$

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Power in an electric circuit

- Power in an electric circuit is **the rate of flow of energy past a given point of the circuit.**
- In **alternating current circuits**, energy storage elements such as inductance and capacitance may **result in periodic reversals of the direction of energy flow.**
- **The portion of power flow that**, averaged over a complete cycle of the AC waveform, **results in net transfer of energy in one direction is known as real power.**

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ESP's motor Power Factor

- The **power factor (PF)** of an AC electric power system is defined as **the ratio of the real electric power** flowing to the load **to the apparent electric power**.
- **Real electric power** or **motor power** is the capacity of the circuit **for performing work**. Often designed as **KW** and measured in kW units, is found by considering that the **real current (I_{real})**:

$$I_{real} = I_{line} \cos \Phi$$

$$KV = (1.732 \times 10^{-3}) \times U_{line} \times I_{line} \cos \Phi$$

- The **apparent electric power** (often designated as KVA and measured in thousands of volt amperes) is the **product of the voltage (U) and current (I)** of the circuit:

$$KVA = (1.732 \times 10^{-3}) \times U_{line} \times I_{line}$$

- Insofar, the **power factor (PF)** will be:

$$PF = KV / KVA = \cos \Phi$$

- **Motor power:**

$$KV = (1.732 \times 10^{-3}) \times U_{line} \times PF$$



ESP install #1. Phillips Petroleum lease, near Burns, Kansas, 1928. Arutunoff is third from right, shown with the original Reda employees.